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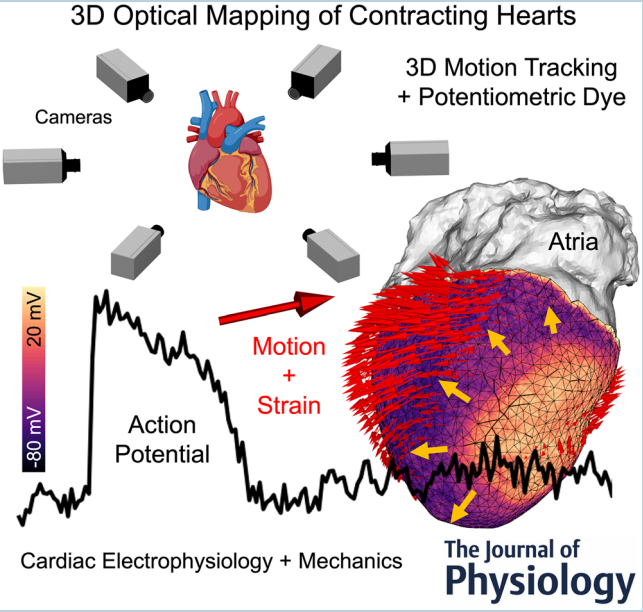
Panoramic voltage-sensitive optical mapping of contracting hearts using cooperative multiview motion tracking with 12 cameras

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Abstact figure legend A panoramic 3D optical mapping system was developed, enabling imaging of action potential waves across the entire strongly deforming ventricular surface of beating isolated hearts. The system comprises 12 high-speed cameras and a soccerball-shaped imaging chamber with 48 light-emitting diodes (LEDs). The LEDs provide pulsed green-blue illumination for ratiometric imaging. Using multiview numerical motion tracking and ratiometric motion artifact compensation, it is possible to measure artifact-free optical traces of action potentials together with tissue strain.

Abstract Voltage-sensitive fluorescence imaging is widely used to image action potential waves in the heart. However, although the electrical waves trigger mechanical contraction, imaging

S. Chowdhary and J. Lebert contributed equally to this work.

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needs to be performed with pharmacologically contraction-inhibited hearts, limiting studies of the coupling between cardiac electrophysiology and tissue mechanics. Here, we introduce a multiple-camera optical mapping system with which we image action potential waves at high resolutions across the entire ventricular surface of the beating and strongly deforming heart. We imaged intact isolated rabbit hearts inside a soccer-ball shaped imaging chamber, facilitating even illumination and panoramic imaging. Using 12 high-speed cameras, ratiometric voltage-sensitive imaging, and three-dimensional (3D) multiview motion tracking, we reconstructed the entire 3D deforming ventricular surface and performed corresponding voltage-sensitive measurements during sinus rhythm, paced rhythm and ventricular fibrillation (VF). Our imaging setup defines a new state-of-the-art in the field and can be used to study the heart's electromechanical physiology during health and disease at unprecedented resolutions. For example, we measured action potential duration and contractile changes in response to pharmacological blockage of potassium ion channels during sinus rhythm, measured electrical activation times and observed mechanical strain waves following electrical activation fronts during pacing, and observed electromechanical vortices during VF.

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Key points

- The heartbeat is controlled by electrical impulse phenomena that trigger contractile motion.
- Optical mapping uses fluorescent dyes to measure electrical impulse phenomena in cardiac muscle tissue. It is an important tool for studying cardiac electrophysiology and rhythm abnormalities.
- With current optical mapping techniques, it is not possible to image beating hearts. Hearts need to be contraction-inhibited using pharmacological agents. This limits optical mapping studies to measurements of electrical activation patterns and action potentials. Tissue strain and contractile motion cannot be assessed simultaneously.
- A novel 3D optical mapping system is presented that enables panoramic imaging of action potential waves across the surface of strongly contracting isolated hearts. With this system, it is possible to measure electrical activation and action potential waves simultaneously with deformation and strain.

Introduction

The heartbeat is a highly dynamic process involving bioelectrical and biomechanical phenomena: each cardiac muscle cell exhibits a sudden, rapid change or depolarization of its transmembrane potential before it contracts. The depolarization is the initial phase of the cardiac action potential, followed by a characteristic plateau phase during which calcium is released into the cell's contractile machinery, producing contractile force and a repolarization phase after which the cell relaxes. On the organ scale, the action potential

Shrey Chowdhary obtained a BSc degree in both Physics and Computer Science at the University of Illinois Urbana-Champaign. He was a Computational Research Specialist at the University of California, San Francisco, where he developed the numerical methods for the present study. Leveraging his interdisciplinary background, he is interested in developing novel computational methods to study chaotic systems. He is currently a PhD student in Physics at the University of Southern California. **Jan Lebert** is a Postdoctoral Scholar at the University of California, San Francisco, working at the intersection of computational cardiology and biomedical imaging. He received his PhD in Biophysics from the University of Göttingen, Germany, for his work on reconstructing cardiac excitation waves from mechanical deformation. Building on his expertise in optical mapping, he investigates the mechanisms of arrhythmias and aims to translate these findings into better patient care.



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spreads from cell to cell, forming an electrical wave that rapidly propagates across the heart muscle, subsequently activating different portions of the heart to contract. Understanding these intricate dynamics is crucial for understanding various forms of heart disease and for developing better diagnostics and therapies to treat them. Although the normal heartbeat is characterized by a rapid depolarization occurring almost everywhere at once via the Purkinje system, abnormal rhythms can be triggered by various electrical activation patterns, such as focal patterns originating from only one location during premature ventricular complexes, or rotating or reentrant patterns during tachyarrhythmias, such as atrial or ventricular fibrillation (VF). More generally, abnormalities in electrical activation can cause contraction abnormalities, and vice versa, as both mutually influence each other. For example, myocardial infarcts are not only associated with stiffer myocardial tissue and reduced contractility, but they can also alter electrical pathways and induce arrhythmias. Moreover, mechanical stretch can influence the action potential via mechano-sensitive ion channels, which, in turn, can affect mechanical contraction, even in healthy tissue. The action potential shape plays an important role in assessing the heartbeat because it not only controls the timing and duration of the heart's contraction via calcium, but also reflects the behaviour of ion channels and the heart's metabolic state and energetics. For example, a prolonged action potential in patients with long-QT syndrome is associated with delayed muscle relaxation, impaired mechanical function and an increased arrhythmia risk as a result of changes in ion channel function that affect the repolarization phase.

For decades, it has been a challenge to image these dynamics throughout the heart, mainly because it requires a high-resolution measurement approach capable of measuring electrophysiology and mechanics simultaneously. Optical techniques employing fluorescence-based measurements are an ideal high-resolution modality for this task. Nevertheless, measuring action potential waves at high resolutions in the beating heart has been a long-standing technical challenge in cardiovascular research (Christoph & Luther, 2018; Christoph et al., 2017, 2018; Garrot et al., 2017; Kappadan et al., 2020, 2023; Lebert et al., 2022; Salama et al., 1987; Zhang et al., 2016, 2023). Electrical mapping techniques, such as multi-electrode arrays or catheter-based electrode mapping (Nash et al., 2006; Narayan & John, 2024), are viable options to map electrical impulse phenomena in the heart and are frequently used in the clinical setting. However, they are associated with low spatial resolution, measure extracellular potentials rather than action potentials, require physical contact with the tissue and generally do not allow simultaneous mechanical measurements of tissue strain. More recently, it became possible to measure electrical impulse phenomena and mechanical stretch simultaneously using flexible nanowire electrode arrays or stretchable patches (Chen et al., 2023; Fullenkamp et al., 2024). However, although these techniques integrate electrical and mechanical measurements and mitigate adverse effects caused by stiffer electrodes, they provide measurement data only from a very limited number of locations and, again, do not measure cardiac action potentials directly, but rather extracellular potentials. Lastly, action potential waves can be measured contactless at high spatial resolutions across the surface of isolated hearts using voltage-sensitive fluorescence imaging or optical mapping (Efimov et al., 2004; Grey et al., 1995; Kay et al., 2004, 2006; Rogers et al., 2007; Salama et al., 1987; Swift et al., 2021). With this technique, the main challenge is that the optical signals arising on the heart's surface are minuscule, and in order to be able to measure action potential waves, it is necessary to combine sensitive optical with precise numerical measurement techniques (Christoph & Luther, 2018; Lebert et al., 2022). Because of its sensitivity, optical mapping is routinely performed in pharmacologically contraction-inhibited hearts (Efimov et al., 2004; Grey et al., 1995; Swift et al., 2021). Voltage-sensitive optical mapping with multiple-camera systems was performed to image action potential waves across the entire heart surface of motion-inhibited hearts (Bray et al., 2000; Kay et al., 2004, 2006; Rogers et al., 2007). Optical mapping of action potential waves on the contracting heart surface is much more challenging than in non-contracting hearts, and has been demonstrated in only a small number of studies (Christoph & Luther, 2018; Christoph et al., 2017, 2018; Garrot et al., 2017; Kappadan et al., 2020; Lebert et al., 2022; Zhang et al., 2016, 2023) and in even fewer studies when performed with multiple-camera optical mapping reconstructing the three-dimensional (3D) deforming heart surface (Christoph et al., 2017; Zhang et al., 2016, 2023).

In the present study, we present the first fully panoramic measurements of action potential waves on the entire strongly contracting ventricular surface of isolated hearts. The 3D measurement data were obtained using a multiple-camera optical mapping system comprising 12 high-speed cameras (Fig. 1, Supporting information, Video SV1). The measurements were enabled by the recent availability of low-cost cameras, a custom-designed spherical imaging chamber, pulsed illumination for ratiometric imaging, and a 3D multiview motion tracking algorith (Figs. 2-5, Supporting information, Videos SV2-SV11). With the spherical, soccer ball-shaped imaging chamber, it is possible to illuminate and image the heart evenly from all sides with up to 48 light-emitting diodes (LEDs) and up to 24 cameras simultaneously. Using 12 cameras, we imaged action potential waves propagating

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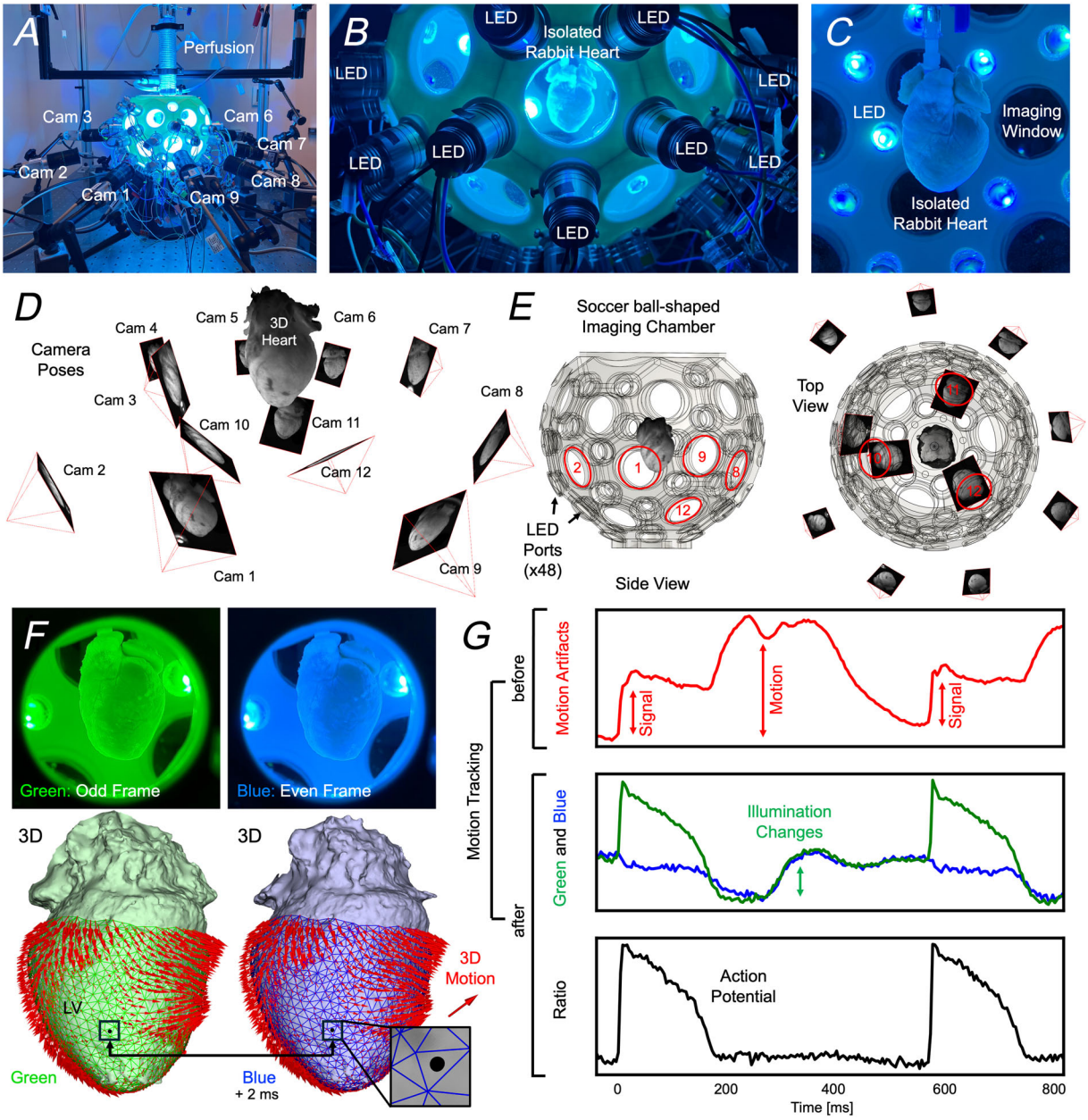


Figure 1. Panoramic voltage-sensitive optical mapping of action potential waves across the entire surface of contracting and strongly deforming isolated hearts
A, spherical 3D-printed soccer ball-shaped imaging chamber with 24 windows, 12 synchronized high-speed video cameras (acA720-520um; Basler) and constant-pressure Langendorff-perfusion system. *B*, view of isolated rabbit heart inside imaging chamber (see Supporting information, Video SV1). *C*, close-up of isolated rabbit heart immersed in Tyrode solution inside imaging chamber. *D*, camera poses (position and orientation) and 3D reconstruction of rabbit heart (not to scale) obtained with 12 cameras (see Supporting information, Video SV2). Imaging was performed at 500 frames s^{-1} with 8 and 12 mm lenses with shorter and longer working distances, respectively. *E*, technical drawing (side and top view) of imaging chamber together with camera poses (right). Horizontal imaging was performed through windows 1–9 and apical imaging through windows 10–12. *F*, continuous green or pulsed green-blue excitation in odd (1, 3, 5, …) and even (2, 4, 6, …) frames, respectively, with up to 48 light-emitting diodes (LEDs) (for more details, see Fig. 5). In pulsed mode, we performed two separate 3D reconstructions using the green and blue video data and subsequently tracked the heart's motion (red vectors, frame-to-frame displacements) over time using the blue video data; for details, see Methods (see Supporting information, Videos SV10 and SV11). Hearts were stained with voltage-sensitive fluorescent dye (Di-4-ANEPPS). *G*, optical traces (normalized dimensionless units) obtained from a 3 × 3 pixel region at the centre of the same triangle (black dot in *F*) before (top, green video) and after 3D motion tracking (centre, green and blue videos), and after both 3D tracking and ratiometric compensation (bottom) (for more examples, see Figs 8*A*, 10 and 11;

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see also Supporting information, Video SV5). While the green trace (obtained with green excitation) carries the signal, the blue trace is neutral and serves as a measurement for local illumination changes that arise as a result of relative motion between the tissue and the LEDs.

across the entire contracting ventricular surface of isolated intact rabbit hearts during sinus rhythm, paced rhythms and VF (Figs. 6-20, Supporting information, Videos SV12-SV19). We mapped action potential waves and measured epicardial deformation and tissue strain at high spatial and temporal resolutions (spatial resolution electrical: 0.5–1.0 million pixels, 120 μm per pixel, spatial resolution mechanical: ∼0.5–3.0 mm per triangle, temporal resolution: 2 ms). These resolutions enabled us, for the first time, to observe electromechanical waves, which are composed of electrical activation fronts and strain waves that closely follow electrical activation throughout the heart. Our imaging system not only comprises the first fully panoramic electromechanical optical mapping system covering the entire 360° ventricular surface of the heart, but also provides unprecedented spatial resolutions and has several key improvements and practical advantages over previously reported 3D electromechanical optical mapping systems (Christoph et al., 2017; Zhang et al., 2016).

Technical background

Optical mapping is a highly sensitive measurement that quantifies small fractional changes in fluorescence exhibited by dyes or genetic indicators in response to a physiological change, such as a change in the transmembrane potential during the cardiac action potential. Motion in cardiac optical mapping studies is typically associated with so-called motion artifacts, which arise even with the slightest motion (Christoph & Luther, 2018; Rohde et al., 2005; Svrcek et al., 2009). With normal heart motion and without numerical motion correction, motion artifacts completely compromise the measurements, prohibiting further analysis of the data. Numerical motion tracking can effectively circumvent motion artifacts (Christoph & Luther, 2018; Christoph et al., 2017, 2018; Kappadan et al., 2020; Khwaounjoo et al., 2015; Lebert et al., 2022; Rodriguez & Nygren, 2015; Zhang et al., 2016) and it is particularly effective when further combined with ratiometric imaging (Kappadan et al., 2020; Zhang et al., 2016). Nevertheless, eliminating motion artifacts and measuring artifact-free action potentials on the contracting heart surface remains a non-trivial challenge that has yet to be solved.

Numerical motion tracking and 3D object reconstruction are routine tasks in the field of computer vision. Yet, despite their widespread use in engineering and other fields, computer vision techniques are still not fully utilized in cardiac optical mapping studies, probably because they require particular technical knowledge, which makes them challenging to adopt by cardiovascular researchers. On the other hand, 3D optical mapping of contracting hearts is a more challenging problem than most other computer vision tasks, which may have also hindered computer vision experts from entering the field. The technical challenges include: the tissue needs to be tracked precisely, at times with subpixel precision, to be able to measure the optical signals without motion artifacts (Christoph & Luther, 2018). The optical signals are small, and even slight tracking errors can lead to large measurement errors. The fluorescence changes rapidly over time, reflecting physiological function, which can irritate tracking algorithms and introduce tracking artifacts (Christoph & Luther, 2018; Lebert et al., 2022). Noise, inhomogeneous illumination and other factors can cause further tracking and measurement artifacts. Overall, the video data comprises very particular image features determined by the fluorescence indicators, filters and low-resolution cameras. The spatial resolution of the videos typically ranges in the order of 100 × 100 to 300 × 300 pixels. Lastly, the heart's motion and deformation are large, potentially causing it to move out of the field of view. Together, these factors make optical mapping of contracting hearts or tissues a uniquely challenging problem. By default and without further customization, stereoscopic imaging or photogrammetry techniques, which are frequently used for shape reconstruction, do not work out-of-the-box and lack the precision required for physiological measurements. Accordingly, attempts to reconstruct and optically map the 3D surface of beating hearts have been reported in only a few studies thus far (Christoph et al., 2017; Zhang et al., 2016, 2023).

Zhang et al. (2016) demonstrated for the first time that action potential waves can be imaged on the contracting 3D left ventricular surface using multiple-camera optical mapping. The imaging, motion tracking, and 3D surface reconstruction were performed with four cameras and required fiducial markers glued to the epicardial surface, which were manually identified and tracked in a semi-automatic fashion through the video images, as described in Bourgeois et al. (2011). Subsequently, the reconstructed 3D heart surface consisted of a low-resolution polygon surface comprising ∼20–30 polygons with vertices defined by the fiducial markers used for tracking. Action potential measurements were averaged within each polygon, without using the full 128 × 128 pixel resolution of the video images. The cameras were positioned at a 45° angle to allow sufficient overlap between their fields of view. This methodology

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was subsequently used to image beating hearts *in vivo* (Zhang et al., 2023).

Christoph et al. (2017) demonstrated that action potential waves can be imaged on the contracting 3D left ventricular surface at higher spatial resolutions and without markers attached to the heart surface. The imaging and 3D reconstruction were also performed with four cameras at 45°, but the heart's motion was tracked automatically in every pixel in each of the four videos, subsequently yielding a high-resolution polygon surface (4000–5000 polygons) with a spatial resolution comparable to the resolution of the video images (128 × 128 pixels). With the four cameras, it was possible to reconstruct almost one-half of the ventricular surface of the beating heart. However, the approach required a static shape reconstruction of the pharmacologically contraction-inhibited heart following each experiment, which is a major limitation. After the optical mapping of the contracting heart was completed, the heart was uncoupled with the uncoupling agent blebbistatin. Once the heart was completely quiescent (15–30 min), it was rotated once around its long axis using a stepping motor while 72 images of its silhouette were captured at 5° intervals over 360°. The silhouettes were then used to reconstruct the 3D geometry of the quiescent heart using a shape-from-contour algorithm. Finally, the 3D mesh of the static heart was deformed using the two-dimensional (2D) motion tracking data obtained from each video camera showing the beating heart. Although this approach provided 3D reconstructions of the deforming heart over time, it could not produce them immediately during imaging. The required additional experimental part at the end with blebbistatin was difficult to execute, added post-processing steps, and introduced potential complications to each experiment that could prohibit further analysis. Most critically, the time available for imaging the beating heart was limited to a few minutes.

In the present study, we address the previous shortcomings by introducing a stereoscopic multiple-camera optical mapping approach that uses multiview imaging and 3D cooperative motion tracking of the heart surface with many low-cost cameras. Our approach neither requires fiducial markers (Zhang et al., 2016), nor a static shape reconstruction obtained with blebbistatin (Christoph et al., 2017), and can be used to immediately and automatically reconstruct the heart surface when imaging action potential waves at high resolutions.

Methods

Ethical approval

All procedures were in accordance with animal welfare law under US law, in accordance with and approved by the Institutional Animal Care and Use Committee (IACUC) of the University of California, San Francisco (UCSF), and followed standard procedures specified by UCSF's Laboratory Animal Resource Centre (LARC). More information about procedures and guidelines can be found online (https://iacuc.ucsf.edu). Animal facilities are run by UCSF LARC under the auspices of the UCSF IACUC, and are staffed by trained husbandry staff, veterinary nurses and veterinarians. UCSF has an approved Assurance of Compliance with US Public Health Service (PHS) Policy on Humane Care and Use of Laboratory Animals (#3400-01) and is accredited by the Association for Assessment and Accreditation of Laboratory Animal Care (AAALAC) (#001084). All animals were treated humanely in accordance with the guidelines of the AAALAC and the National Institutes of Health (NIH). All procedures were covered under protocols AN193503 and AN206877 with the title 'High-resolution *ex vivo* imaging of isolated rabbit and pig hearts' valid from 6 April 2022, until 23 February 2028. All procedures and the treatment of the animals comply with the Journal of Physiology's ethical principles.

Heart isolation procedure

This study only involved tissue collection. No procedures were performed on living animals. No animals were excluded and no unexpected events occurred. The only selection criterion was age. Healthy New Zealand White rabbits ($n = 9$, sex: 8/1 female/male, aged 12–24 months; Western Oregon Rabbit Co., Philomath, OR, USA) were deeply anaesthesized using buprenorphine (0.01–0.05 mg kg^{-1}, I.M. injection), isofluorane (1%–5% inhalant to effect for anaesthesia maintenance, high dose, 15–30 min) and ketamine (35–50 mg kg^{-1}, I.M. injection) plus xylazene (5–10 mg/kg, intramuscular injection) or medetomidine (0.5 mg kg^{-1}, I.M. injection). Buprenorphine was administered 30 min prior to the administration of ketamine/xylazine/medetomidine. Hearts were excised quickly within 10–15 min during open-chest surgery and immediately transferred into ice-cold cardioplegic solution. All procedures were performed by trained LARC staff. As a result of the deep anaesthesia and excision of the heart, rabbits were killed without feeling pain or distress. Hearts were subsequently prepared for retrograde Langendorff perfusion and transferred into the imaging setup within 20–45 min.

Langendorff perfusion of isolated hearts

Hearts were placed inside an imaging chamber filled with warm, oxygenated Tyrode solution (5 L) and attached to retrograde Langendorff perfusion within 20–45 min after excision. Hearts were perfused in constant-pressure

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mode. The Langendorff perfusion system comprised a 5 L water-jacketed reservoir (model no. 120142-5; Radnoti, Monrovia, CA, USA) that was positioned about 0.6–0.7 m above the heart and connected via tubing to a heating coil with integrated bubble trap (model no. 158840; 10 mL; Radnoti) positioned directly above the heart. The perfusion pressure was regulated by adjusting the height of the water reservoir to create perfusion pressures of 50 ± 5 mmHg. Perfusion pressure was monitored using a digital manometer (HT-1890; Hti-Xintai Instrument, Dongguan City, China). The Tyrode was oxygenated and its pH level was kept constant at 7.4 by bubbling carbogen into the reservoir throughout the experiment. The pH level was monitored using a digital pH meter (PH220; Extech, Taipei City, Taiwan). An immersion circulator-heater (Polystat; Cole-Palmer, Vernon Hills, IL, USA) was used to pre-heat water to $37 \pm 0.5^{\circ}$C, which was then pumped through the water-jacketed reservoir and heating coil to pre-heat the Tyrode solution. The resulting temperature of the Tyrode flowing out of the heating coil into the heart was $36.75 \pm 0.25^{\circ}$C. The Tyrode solution inside the imaging chamber was continuously exchanged with pre-heated Tyrode that was pumped out of and recirculated back into the chamber to maintain a constant Tyrode temperature of $36.0 \pm 0.5^{\circ}$C. The Tyrode solution was prepared in batches of 5 L of deionized water with 33.6 g of sodium chloride (115 mM), 1.75 g of potassium chloride (4.7 mM), 0.71 g of magnesium chloride (1.5 mM), 1.1 g of calcium chloride (2.0 mM), 0.3 g of sodium phosphate (0.5 mM), 11.75 g of sodium bicarbonate (28 mM) and 18.0 g of glucose (20 mM). For some experiments, we prepared low-potassium concentration Tyrode with 0.875 g of potassium chloride (2.35 mM), 0.052 g of barium chloride (50 μM) and 0.194 g of caffeine (0.2 mM), as described in Myles et al. (2015).

Soccer ball-shaped imaging chamber

We designed a spherical soccer ball-shaped imaging chamber, which facilitates panoramic optical mapping of contracting hearts with up to 24 cameras simultaneously (Figs. 1 and 2A). The chamber allows a dense coverage of the heart during imaging with overlapping fields of views (Fig. 3*A* and *B*), at the same time as providing an evenly distributed, unobstructed illumination of the heart through additional dedicated openings with up to 48 LEDs (Figs 1*A*–*C*, *E* and 5*A*–*C*). The chamber's geometry is based on a truncated icosahedron or soccer ball geometry. The soccer ball is open at the top, has 24 penta- or hexagonal surfaces and an outflow at the bottom. Imaging can be performed through round windows in each of the 24 surfaces. The angle of the optical axes between two adjacent windows is 37.4° or 41.8°, depending on whether the angle is measured between the penta- and/or hexagonal surfaces. Illumination can be provided through 48 small round windows (LED ports) located at the vertices between the penta- or hexagonal surfaces. The LEDs are mounted inside these LED ports. The size of the chamber is determined by the working distance of the cameras and lenses, the size of the heart and the size of the windows through which imaging is performed. The volume of the chamber is ±5 L. Hearts are positioned at the centre of the imaging chamber and can be imaged equally through all windows without obstructions. Similarly, the heart can be illuminated by all LEDs without obstructions. Both the 24 large round window openings for imaging and the 48 small openings for the LEDs are covered by round plexiglass windows. The plexiglas windows are glued into the window openings using silicone. Altogether, the imaging chamber has 72 window openings (large and small). The chamber shown in Figs 1, 2*A*, 3*C* and 5*A*,*B* was 3D printed (3D printer: CR-10 V2; Creality, Shenzhen, China; material: white PLA+; Overture, Missouri City, TX, USA).

Multiple-camera optical mapping

Multiple -camera optical mapping was performed using 12 high-speed complementary metal-oxide-semi-conductor (CMOS) cameras (acA72-0520um, Basler, Ahrensburg, Germany), which were positioned around the imaging chamber on an optical table (TMC, Peabody, MA, USA) (Figs 1*A*,*D* and 2*B*; see also Supporting information, Video SV2). The cameras were aligned and fixed in place using articulator arms (Noga, Kanazawa, Japan). The average angle between neighbouring cameras was $\sim 40 \pm 10°$, as shown in Figs 1 and 3*C*. With 12 cameras, it is possible to achieve a sufficiently large overlap between the fields of view of the cameras (Fig. 3*A*) and dense coverage of the ventricular surface with at least three cameras observing one point simultaneously across the surface (Fig. 3*B*). The cameras produce 12-bit (dynamic range: 0–4096 intensity counts) greyscale video images with a maximal spatial resolution of 720 × 540 pixels. Imaging was performed at speeds of 500 frames s^{-1} with a spatial resolution of 440 × 320 pixels. The cameras were triggered using a signal generator (FY8300; FeelElec, Zhengzhou, China) and custom software to synchronize the video acquisition. Emission light was collected through machine vision lenses (3x CF8ZA-1s from below, 9x CF12ZA-1s horizontal; Fujinon, Tokyo, Japan) mounted onto each of the cameras together with emission filters, see next section. With the magnification of the lenses, the video images covered a field of view of ∼4 × 5 cm at working distances of ∼10–15 cm (Fig. 1*A* and *D*). The videos from the cameras were streamed via USB onto a computer (AMD Ryzen Threadripper 3960X CPU, AMD; Santa Clara, CA, USA; 2TB SSD working memory; TRX40 mainboard; Gigabyte Technology Co., New Taipei

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City, Taiwan) using the nine integrated USB ports and additional external USB hubs (10 Gbps, USB 3.2 Gen 2).

Ratiometric voltage-sensitive optical mapping

Hearts were stained with two 250 µL bolus injections of voltage-sensitive fluorescent dye (Di-4-ANEPPS; Biotium, Hamburg, Germany; 5 mg dissolved in 5 mL of dimethyl sulfoxide, resulting in a stock solution with a concentration of 2.078 mM) into the bubble trap. The second bolus was injected 5 min after the first bolus and imaging was started ∼10–15 m in after the first injection. In this study, all hearts were imaged without pharmacological uncoupling agents, such as blebbistatin. Optical mapping was performed either in continuous mode with continuous green excitation light or in ratiometric mode with pulsed green and blue excitation light (Bachtel et al., 2011; Bourgeois et al., 2011;

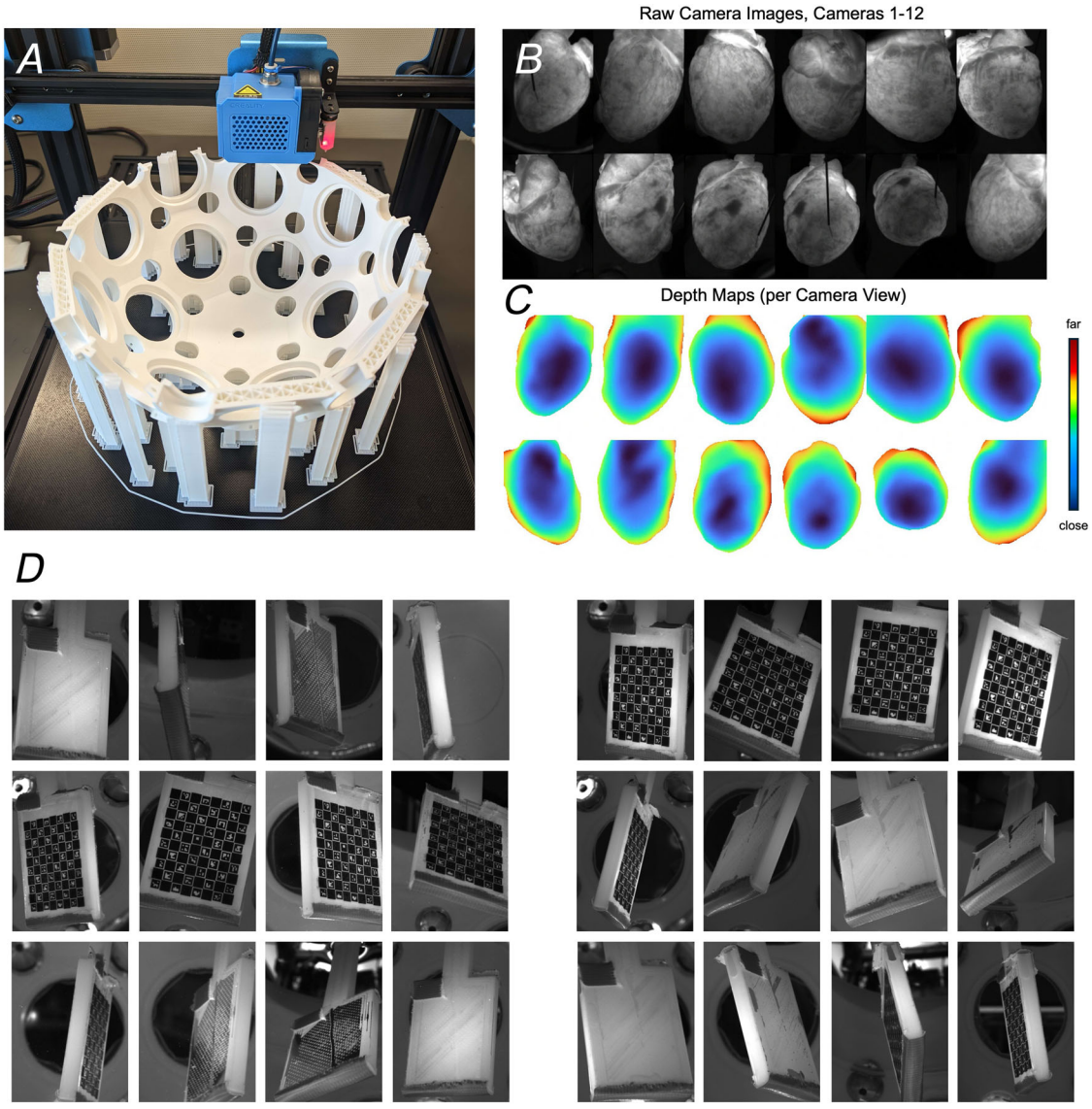


Figure 2. Panoramic optical mapping with up to 24 cameras
A, 3D printing of soccer ball-shaped imaging chamber (Fig. 1). The chamber has 24 large windows for imaging and 48 small windows/ports for LEDs. *B*, raw greyscale camera images from cameras 1–12 showing isolated rabbit heart during voltage-sensitive optical mapping with continuous green illumination. In this study, we refer to each set of (12) camera images as one frame. *C*, corresponding depth maps indicating the distance of the heart surface to the camera (dark blue: close, red: far). The information can be obtained from the three-dimensional reconstruction and multiple-camera tracking of the heart surface. *D*, calibration target with ChArUco pattern inside imaging chamber as seen by 12 cameras for two different poses of the target. At least 36 different sets of images were taken such that the target was visible at least three times in each camera at the same time as being visible in two to three of the neighbouring cameras.

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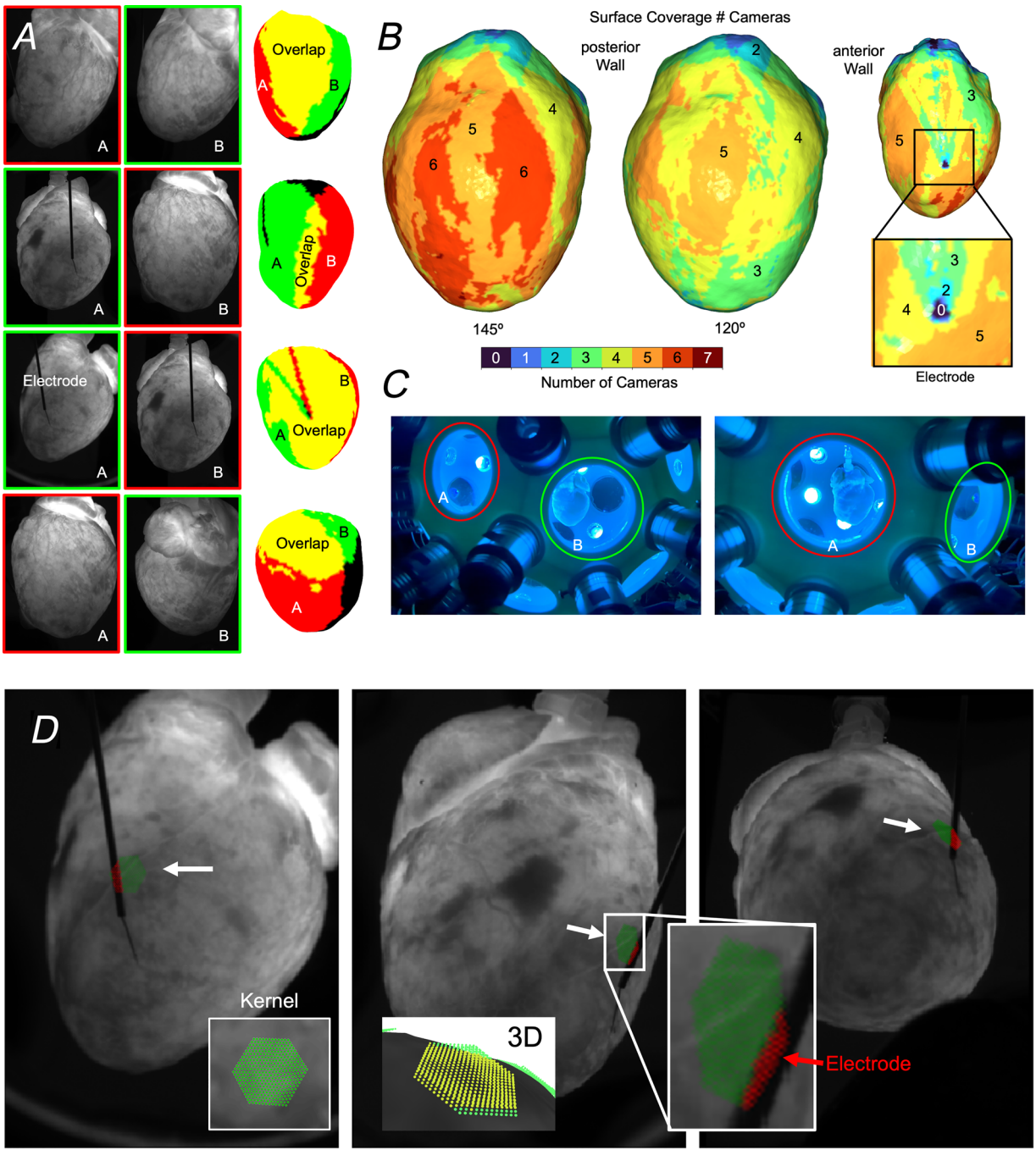


Figure 3. Coverage of 3D heart surface and multiview tracking with occlusions
Coverage of heart surface with 12 cameras configured as in Fig. 1*E*. *A*, camera images from two adjacent cameras (here two random pairs A&B indicated in red/green) and overlap (yellow). Tissue covered by the pacing electrode in one camera is visible in the other camera. *B*, coverage of tissue surface by number (#) of cameras. The tissue is always covered by at least four to six (left) or three to five cameras (centre, right) when allowing camera normals within angles of 145° or 120° around the local surface normal, respectively. The tissue behind the pacing electrode can always be seen by at least three cameras, except for a small area underneath the tip of the electrode (right). *C*, view through two adjacent windows into imaging chamber. The average angle between cameras is about 35–45°. *D*, tracking algorithm ignores pacing electrode. The mesh tracking kernel samples image points from the ventricular surface (green) using a 2D plane that is an approximation of the surface and is aligned with the centre vertex normal (3D). With our extensions to the method developed by Klaudiny and Hilton (2011), the kernel ignores tissue that is occluded by the pacing electrode (red). The tissue behind the electrode can be tracked nevertheless, because it is in almost all cases visible in multiple other camera views; see also (*A*) and (*B*).

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Kappadan et al., 2020). We used up to 48 high-power, multicolour RGBW (red–green–blue–white) LEDs to illuminate the ventricles from all sides (Cree XML RGBW Star LED; Cree LED, Durham, NC, USA) (Figs 1*A*,*F* and 5*A*–*C*). The LEDs were positioned in 48 ports between the imaging windows and directed at the centre of the imaging chamber. In continuous green mode, we mounted collimator lenses (20 mm diameter poly-carbonate, wide spot pattern, Carclo) in front of the LEDs. In ratiometric mode, we performed some experiments with collimator lenses (Figs 1*G*, 5*G*, 7*C*,*D*, 9*E*, 10, 12*D*,*E*, 13*A* and 14) and some without collimator lenses (Figs 8, 11 and 13*B*). Without collimator lenses, the heart appears to be illuminated more evenly, both across the surface as well as across the green and blue channels (Fig. 5*E*–*H*). Each LED comprises four diodes, a red, green, blue and white diode, which are positioned next to each other in close proximity and can be powered individually (Fig. 5*C*). In continuous mode, we powered only the green diodes continuously. In ratiometric mode, we powered the blue and green diodes alternatingly with two power sources, rapidly switching the power on and off between the blue and green sets ($2 \times 48 = 96$ total) using a custom-made LED driver. We used two programmable direct current power supplies (KWR103, 60V/15A; Korad Technology, Dongguan City, China) as power sources for the LEDs. The triggering of the LED driver was synchronized with the camera acquisition, such that odd and even video frames were illuminated with blue or green excitation light for 2 ms, respectively (Figs 1*F* and 5*D* and *E*; see also Supporting information, Video SV9). Correspondingly, the effective acquisition speed of the blue and green videos was 250 frames s^{-1} (Fig. 20). In continuous mode, we used red longpass emission filters (Dark Red #29; Heliopan, Gräfelfing, Germany), which block green excitation light and transmit red and near-infrared fluorescent light. In ratiometric or pulsed mode, we used orange-red band-pass filters (605-70; Chroma, Foothill Ranch, CA, USA) to create a negative fractional change in fluorescence $\Delta F/F$ under green illumination and no fractional change under blue illumination. In this configuration, the blue channel can be used to measure changes in the illumination on the heart surface as it moves through an inhomogeneously illuminated scene. During ratiometric imaging, the blue and green excitation light from the LEDs was filtered using shortpass filters (550 nm cut-off wavelength, model FESH0550; Thorlabs Inc., Newton, NJ, USA).

Rhythm control

Paced rhythms were induced using a custom-made micro-electrode (FHC Inc., Bowdoin, ME, USA) (Figs 2*B* and 3*D*) connected to an isolated pulse stimulator (model 2100; A-M Systems, Carlsborg, WA, USA), which was externally triggered using custom-made software. Regular ventricular pacing was induced using 8 ms long biphasic pulses applied at 3–8 Hz. VF was induced using burst pacing at frequencies ranging from 15 to 50 Hz and terminated using a custom-made defibrillator. Action potential prolongation was caused by switching to low-potassium (50% concentration) Tyrode solution containing barium-chloride, as described in Myles et al. (2015). The electrocardiogram and camera triggers were recorded during the experiment using an analog-to-digital data acquisition system (PowerLab 16/35; ADInstruments, Bella Vista, NSW, Australia).

Camera calibration

Camera calibration refers to the process of determining extrinsic and intrinsic camera parameters with the goal of undistorting the cameras and establishing the relative positioning and alignment of the cameras with respect to each other. We used multical to perform the camera calibrations (Oliver Batchelor, 2023). Camera calibration was performed after each imaging experiment using an 8×10 ChArUco calibration target with a 4 mm checkerboard pattern and ArUco fiducial markers in each of the black squares (Garrido-Jurado et al., 2014) (Fig. 2*D*). We manually positioned and rotated the calibration target within the imaging chamber, at the same time as taking images with all cameras simultaneously for each pose of the target (e.g. 12 images if we used 12 cameras), which we refer to as one image set. More specifically, we focused on one camera at a time and ensured that the target was visible in that camera as well as in as many neighbouring cameras, as possible, during thre acquisition of at least two or three image sets (e.g. $2–3 \times 12$ images if 12 cameras were used). For each pose, the target was typically observable by at least two to four cameras simultaneously. The calibration procedure typically required 30–100 image sets in total (or ∼400–1200 images in total). We added image sets until multical succeeded in completing the calibration procedure, and indicated sufficient overlap and detections of corresponding features of the calibration target between all camera pairs. With both the extrinsic and intrinsic camera parameters, we are able to transform a 3D world co-ordinate to an image co-ordinate, but, for the transformation to be accurate, we must account for camera distortion. In addition to the camera's inherent distortion, the refraction from the light passing from water to glass to air introduces additional distortion. Kwon (1999) showed that the refraction error can be accounted for using standard camera distortion calibration.

3D motion tracking and surface reconstruction using cooperative multiview tracking

Using the camera calibration, we used COLMAP (Schönberger et al., 2016) to produce raw, static 3D

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reconstructions of the heart for each frame over time (Figs 4*A* and 20). A frame corresponds to a set of video images acquired at the same time by the different cameras (Fig. 2*B*). Using the 3D surface reconstructions, it is possible to create depth-maps (Fig. 2*C*). The static reconstructions were performed independently from each other frame-by-frame over the sequence of frames, producing a sequence of unique, uncorrelated raw meshes with different numbers of vertices and polygon faces (see Supporting information, Video SV4). Visually, these meshes reflect the motion of the heart when texturized, but they cannot be used to measure action potential waves or strain as the numbers of vertices and faces fluctuate over time. The topology of each mesh is different and the vertices and faces are uncorrelated over time across meshes.

To compute a single moving mesh that represents a dynamic reconstruction of the moving heart surface, we used the 3D mesh tracking technique described by Klaudiny and Hilton (2011) (Fig. 4*F*). This cooperative patch-based mesh tracking technique produces a single moving mesh with fixed topology and the same number of vertices over time from the individual meshes from the static reconstructions. It does so by tracking the vertices of a template mesh (Fig. 4*B* and *F*), searching for the corresponding tissue segments in both the other already pre-computed raw meshes and in the different camera images of another frame. In our case, the template mesh is derived from one of the raw meshes obtained during the static reconstructions which then serves as the reference frame (typically the first frame). The template meshes were created manually by cutting off the atria around the base of the heart from the selected raw mesh using Blender (Blender-Online-Community, 2018). Next, a sequence of remeshing, decimating and smoothing (Taubin filtering) was applied to the template mesh using Blender (Blender-Online-Community, 2018), VTK (Schroeder et al., 2006) and Open3D (Zhou et al., 2018). The remeshing and decimating methods and their input parameters [e.g. vtkQuadricClustering(input parameter)] determine the spatial resolution of the mesh and subsequently of the tracking and displacement vector fields. Based on this preprocessing, the resulting template meshes consist of ∼1000–5000 vertices and about twice as many triangular faces (Fig. 4*B*; see also Supporting information, Video SV4). At these resolutions, we found the triangles to be ideal in size (neither too small, such that we could still map sufficiently many pixels on each triangle, nor too large). Note that the meshes displaying strain are coarser than the meshes displaying electrical activation. We chose lower resolutions when computing strains to facilitate smoother strain patterns. Depending on the input parameters of the raw mesh reconstruction, the number of vertices or faces of the raw static meshes is in the order of tens to hundreds of thousands of vertices/faces (10^4–10^5). In our experience, the exact number is not critical as long as the resolution is around two to five times higher than the number of vertices/faces of the tracking mesh. The raw meshes also do not need to include (grey-) intensity values associated with the vertices (see Supporting information, Video SV3). The reference frame shows either the heart shortly before the depolarization of the ventricles during sinus rhythm or is an arbitrary frame in case of arrhythmias. To produce the next mesh in the sequence of frames, the algorithm uses first the reference mesh and then each previous mesh as a starting point and then computes and minimizes an error function for each vertex v_e using a patch-match inspired approach (Barnes et al., 2009). This results in a new time-varying mesh with vertices moving in accordance with the video data from the different cameras.

In more detail, the vertex error function is comprised of an image-based component and a mesh-based component. For the image component, we created a set of sample points surrounding the vertex on the mesh's surface, as shown in fig. 1 in Klaudiny and Hilton (2011), Fig. 4*E* (illustration) and Fig. 3*D*. This is carried out for the vertex in the previous mesh and vertex in the new mesh, which is initially a replication of the previous mesh. These sample points are projected into the camera images in which they are visible. The pixel values of the previous mesh's vertex sample points are compared with the new mesh's vertex sample points using a normalized cross-correlation similarity measure. This ensures that the area around the vertex in the new mesh is similar to the corresponding vertex in the previous mesh. For the mesh component, it computes the distance of the vertex from the surface of the raw, static mesh. This distance is passed through the Tukey biweight function to reduce the effect of outliers (Klaudiny & Hilton, 2011). The image error i_e and mesh error m_e components are summed with a weighting term w for the mesh error: $v_e = i_e + w \cdot m_e$. The mesh error reduces the search space and drift over time. We optimized the vertex error using a cooperative optimization algorithm that optimizes all vertices together (Barnes et al., 2009). It begins by randomly translating each vertex multiple times and then keeping track of the displacement that produces the lowest error. In the next step, each vertex applies its neighbouring vertices' best displacements to itself and checks whether it improves the error. The random search and neighbour searching stages are repeated in succession four to five times to find the vertex positions for the new mesh that minimize the error. Subsequently, we take the displacements of the new mesh's vertices with respect to the previous mesh and apply a weighted smoothing to the displacements based on the vertex error (Klaudiny & Hilton, 2011). This results in a completed mesh tracking step for one frame, which is then repeated sequentially for all frames to produce the time-varying mesh over the entire sequence of frames.

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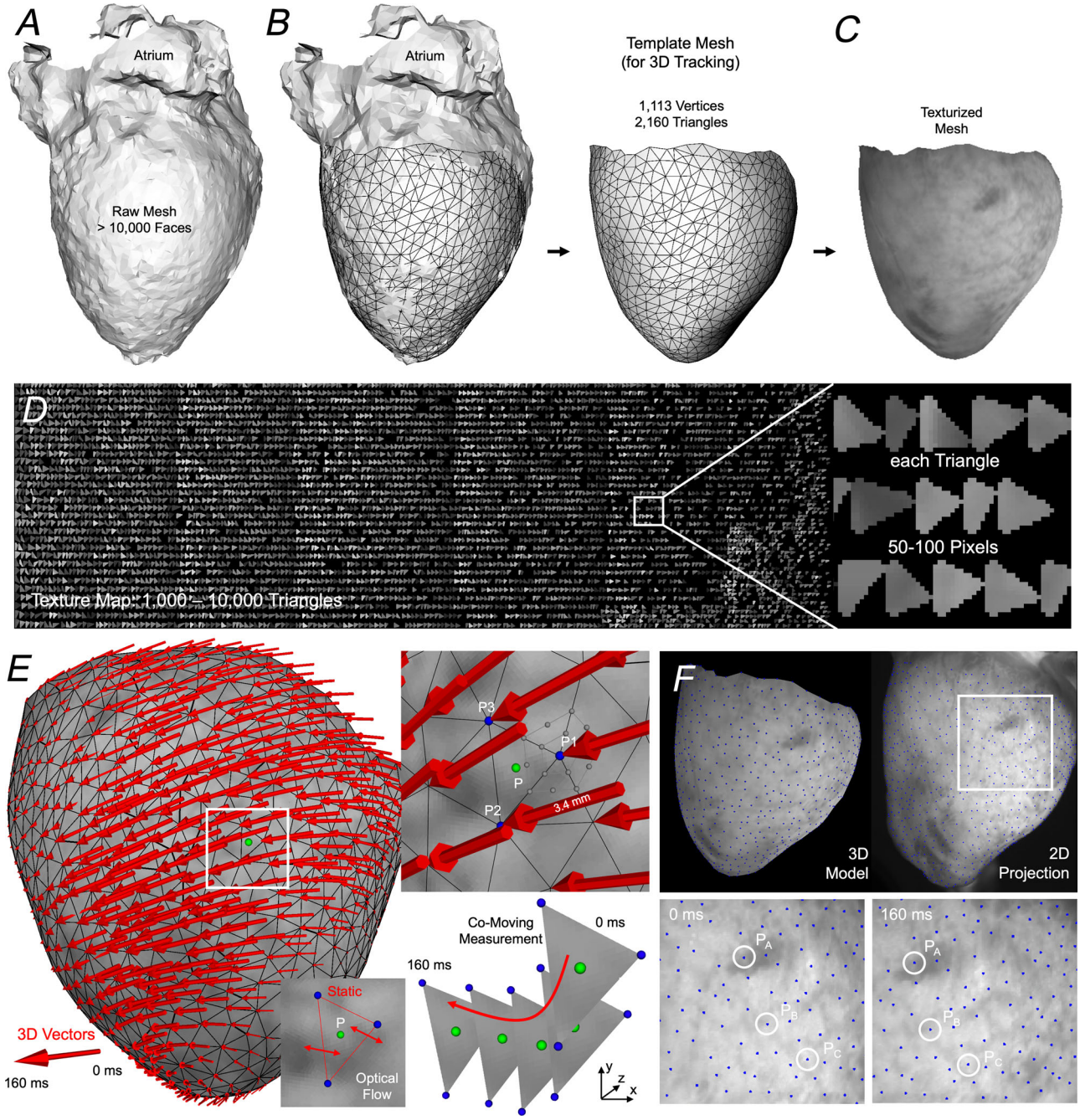


Figure 4. Panoramic 3D reconstruction, tracking, and voltage-sensitive optical mapping of moving heart using multiview cooperative numerical motion tracking with 12 calibrated cameras
A, high-resolution raw mesh resulting from static 3D reconstruction (photogrammetry-like technique or patch-matching with COLMAP) showing 3D rabbit heart surface during systole (160 ms). The motion of the heart is initially represented by a sequence of independent raw meshes which each consist of an uncorrelated number of vertices and faces (see Supporting information, Video SV4). *B*, the motion is tracked over time using a template mesh (black wireframe mesh), which typically consists of ∼2000–10,000 (triangular) polygon faces. 3D tracking is performed using a multiview vertex-based mesh tracking approach developed by Klaudiny and Hilton (2011); see also (*E*). *C*, after tracking, the moving mesh is texturized with the greyscale video data averaging data from multiple cameras (see Supporting information, Video SV2). *D*, greyscale texture map containing thousands of triangles used to texturize the mesh in (*C*). The ventricular surface is resolved by ∼0.5–1.0 million pixels, as each triangle contains ∼50–150 pixels, at a spatial resolution of ∼120 µm per pixel. The pixels are superpositions of pixels from multiple cameras (Fig. 3*A* and *B*). *E*, displacement vectors (red) indicating shifts of epicardial surface from diastole (0 ms,

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shortly before depolarization) to systole (160 ms). Close-up: tracking is performed per vertex (here shown for vertex P1): a mesh-based kernel consisting of a set of sample points (grey, illustration) surrounding each vertex is projected into and used to identify tissue movements in each camera; for details, see Klaudiny and Hilton (2011) (see Fig. 3*D*). *F*, mesh vertices (blue points) superimposed on texturized mesh (3D model) and projected into one of the camera images (2D). The projected vertices follow the motion of the heart in each individual video (e.g. points P_A, P_B, P_C) (see Supporting information, Video SV6).

In continuous or single channel mode with green-only illumination, we performed tracking from frame to frame (2 ms temporal resolution) using a template mesh created from the first raw mesh. In ratiometric or pulsed mode, we created a template mesh from the first raw mesh in the blue channel, and then used this mesh to track the motion separately in the blue and green channels from frame to frame (4 ms temporal resolution) (Fig. 20). Tracking fluorescence videos with substantial temporal fractional change in intensity (e.g. $\Delta F/F \approx 10\%$) may yield tracking artifacts in that the tracking algorithm inadvertently tracks intensity changes caused by the action potential rather than motion, as described in Christoph and Luther (2018) and Lebert et al. (2022). This issue potentially only arises during continuous green imaging or in the green channel during ratiometric imaging. By contrast, such tracking artifacts do not arise in the blue channel, as there are no action potential signals in this channel. To avoid tracking artifacts, we tracked contrast-enhanced videos, as described in Christoph and Luther (2018), in continuous green mode or the blue channel in ratiometric mode. Ultimately, in ratiometric mode, we used the blue mesh obtained by tracking the blue videos to perform all post-processing, such as texturizing the meshes and extracting signals, and applied the same post-processing steps to both the blue and green video data. The two time-varying meshes resulting from tracking the blue and green channels equally reproduce the motion of the ventricular surface (see Supporting information, Videos SV10 and SV12). Independently, we found that our mesh tracking methodology does not appear to be susceptible to such artifacts. The data shown in Fig. 16*I*–*K* validate that the 3D tracking is not sensitive to or compromised by the fluorescence signal in the green channel. It compares the tracking output with and without contrast-enhancement (Christoph & Luther, 2018; Lebert et al., 2022). Contrast-enhancement is a technique that reduces tracking artifacts by suppressing fluorescence signals during pre-processing of the video data. Nevertheless, tracking with or without contrast-enhancement yields almost identical strain-rate patterns, which is an indication that the 3D mesh tracking is not sensitive to the fluorescence signal. The insensitivity of our tracking method to the signal in the green channel is presumably facilitated by the normalized cross-correlation similarity measure in the image-based part of the error function, which suppresses temporal changes in the image intensity.

Post-processing

The vertices of the time-varying mesh describe the motion of the heart surface. With that information, signal analysis can be performed in a co-moving frame of reference (Fig. 4*E*). Accordingly, we warped the video images using the 2D projections of the movements of the 3D mesh in the camera images to be able to further process the data in an image-based co-moving frame of reference (see Supporting information, Videos SV7 and SV8). This step significantly reduced motion artifacts. Because the resolution of the mesh is lower than the resolution of the video images, we further performed 2D optical flow-based motion tracking as described in Lebert et al. (2022) using a GPU-accelerated version of the Farnebäck algorithm (Farnebäck, 2003) on the already warped video images and warped them further with respect to the reference video images (see Supporting information, Video SV8). The 2D tracking and warping was performed using optimap (Lebert & Christoph, 2023). This last step significantly reduced residual motion and motion artifacts in the triangles between the vertices (Fig. 9*C*). Accordingly, all motion-compensated optical maps shown throughout this study were compensated first using the 3D tracking and then 2D tracking. Otherwise, we texture-mapped the raw greyscale video images showing the moving heart onto the moving mesh to produce texturized meshes as shown in Figs 4*C*,*E*, 6*B*, 7*A* and 18*A* (see also Supporting information (Videos SV2, SV4, SV11, SV12 and SV18). Similarly, we projected the warped, processed and then unwarped videos onto the moving mesh. Projections were performed using a weighted average based on the angle between the mesh's surface normal and the camera's direction. The texture data were saved in 8-bit format as a PNG file so the textures could be processed by general mesh viewing tools. The average angle between cameras was $54 \pm 18^\circ$, which includes nearest- and second-nearest-neighbouring cameras, as well as cameras further away (as some windows were not used for imaging). Each camera overlaps with two to four other cameras (Fig. 3*B*).

All post-processing was performed in a co-moving frame of reference and in an image-based fashion in the individual warped video images using optimap (Lebert & Christoph, 2023). The 3D post-processed (e.g. pixel-wise normalized) optical maps are the average of multiple post-processed videos and are combined using a weighted average with the weights being the angle between the

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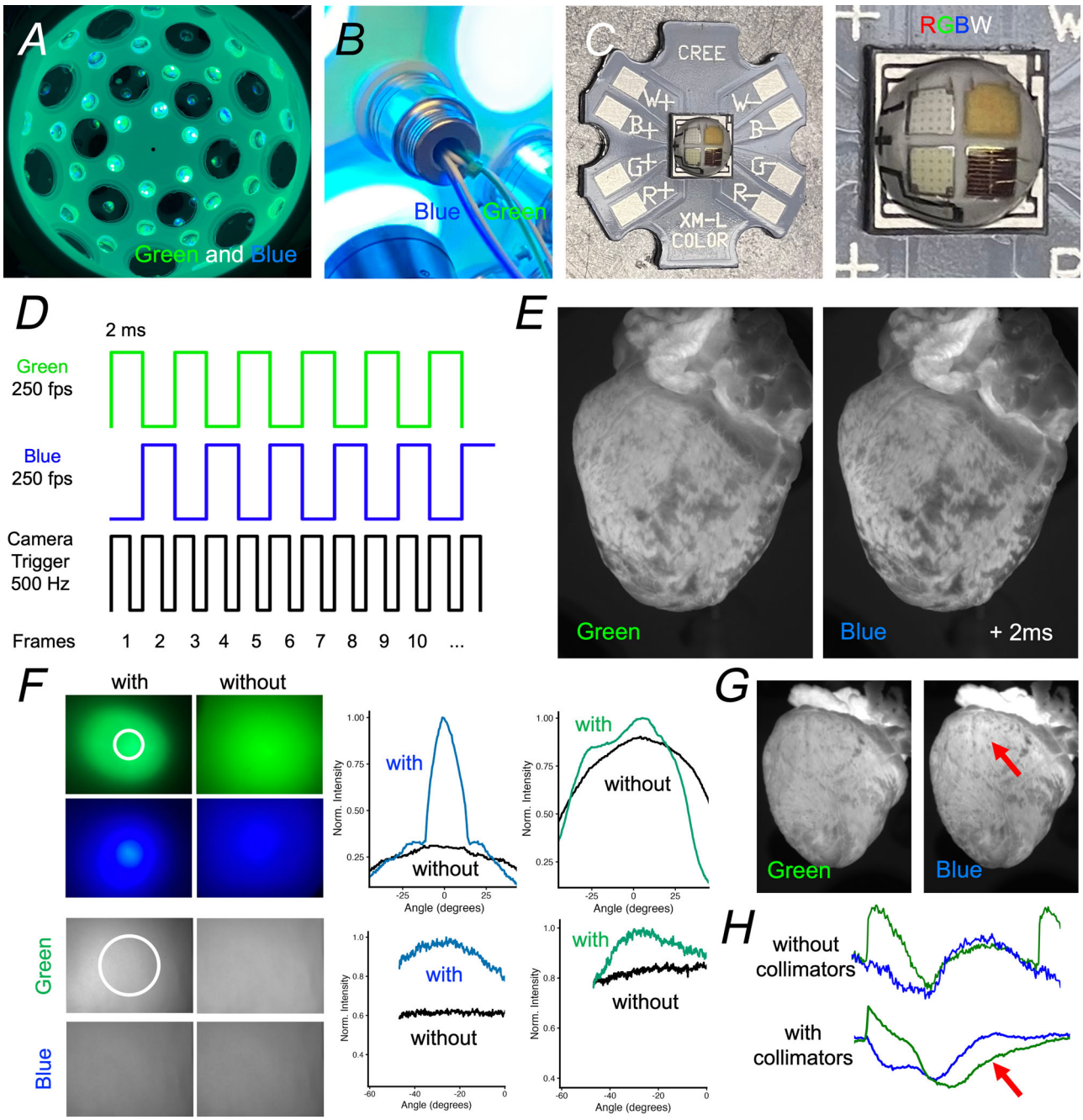


Figure 5. Ratiometric 3D optical mapping with pulsed green and blue excitation
Panoramic pulsed excitation for voltage-sensitive optical mapping with 3D motion tracking and ratiometric motion artifact compensation. *A*, interior of imaging chamber under simultaneous green and blue illumination. *B*, single LED mounted onto heatsink and inserted into LED port of imaging chamber with green and blue wiring for the two green and blue diodes, respectively. *C*, red–green–blue–white (RGBW) LED (Cree XML RGBW Star LED) with adjacent diodes which can be powered individually. *D*, triggering scheme for pulsed green-blue illumination and video frame acquisition. The switching back and forth, alternatingly powering the green and blue channel, is synchronized with the video acquisition (500 frames s^{-1}) yielding two videos at half the video acquisition rate (2 × 250 frames s^{-1}). The pulsed illumination and acquisition are performed simultaneously with all 12 cameras to create two 3D meshes (see Fig. 1*F*; see also Supporting information, Videos SV10 and SV11). *E*, blue and

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green video frames acquired 2 ms apart in alternating sequence. Excitation without collimator lenses. The signal reporting the action potential wave is only present in the green channel (see Supporting information, Video SV9). *F*, excitation light from one LED at a distance of 0.5 m with and without collimator lenses (see Methods), shown in photo, greyscale video image and beam profile. White circle: approximate size of heart. *G*, blue and green frames acquired with collimator lenses. The illumination patterns exhibit differences (red arrow, particularly strong example). *H*, effect of collimator lenses onto green and blue traces. With collimators, the traces deviate more often from each other (see Figs 10 and 11).

triangle normals and camera's view direction. To visualize action potential waves (see Supporting information, Videos SV12, SV14 and SV18), we normalized the optical signals representing transmembrane potential changes in the warped (both ratiometric or non-ratiometric) video data per pixel over time (Lebert & Christoph, 2023). We used a temporal pixel-wise normalization (optionally with a sliding-window normalization, 30–50 frames length) or a frame-wise difference computation with subsequent temporal (sliding-window) normalization to amplify the optical signal (Figs 7–20). Spatial or temporal smoothing of the optical maps was not applied throughout this study. Lastly, we texture-mapped the post-processed pixel-wise normalized optical maps onto the 3D heart surface after rewarping them back into the static laboratory frame.

Action potential duration measurement

To calculate action potential durations (APD) and APD maps, we extracted raw ratiometric optical traces from the centre of each triangle (averaging from 3×3 or 7×7 pixels) (see Supporting information, Video SV5) and further post-processed the traces using a custom detrending and baseline removal algorithm (Fig. 14). Action potential upstrokes were computed by detecting peaks in the temporal derivative of each optical trace. Next, smoothed versions of the first and second temporal derivatives were calculated using Savitzky–Golay filtering, and negative peaks in the first and positive peaks in the second derivative were calculated as estimators for the repolarization phase and end of the repolarization, respectively (Fig. 14*A*). The peak detections were only performed between 100 and 300 ms after each upstroke. Detections yielding fewer or more than anticipated peaks were excluded from further analysis. Subsequently, the parts of the trace between the upstrokes and repolarizations were removed and missing values were linearly interpolated. The remaining baseline was smoothed using Savitzky–Golay filtering and then subtracted from the original trace yielding traces as shown in Fig. 14*B*. APDs were calculated for each action potential individually and averaged per trace if multiple action potentials were present (Figs 12*E* and 13*A*). Traces with two APs and substantial mismatches in APD between two subsequent action potentials were excluded from further analysis. APD maps were obtained by repeating the APD measurement for every triangle.

Phase maps and phase singularity calculation

Phase maps were computed using optimap (Lebert & Christoph, 2023) from the motion-stabilized and pixel-wise normalized optical mapping videos in each camera individually. The phase angle was computed for each time-series using the Hilbert transform as described in Iyer and Grey (2001). The resulting phase maps were smoothed using a complex order parameter filter as described in Lebert et al. (2021). The smoothed 2D phase maps were then projected onto the 3D mesh and averaged from multiple camera perspectives if the respective pixels overlapped. Phase singularities were computed on the 3D mesh surface by computing the circular integral of the gradient of the phase, as described in Li et al. (2020).

Results

We performed fully panoramic voltage-sensitive optical mapping with contracting isolated rabbit hearts, imaging action potential waves across the entire 3D contracting ventricular surface during sinus rhythm, ventricular pacing and VF (Figs 6–19; see also Supporting information, Videos SV2–SV3 and SV12–SV19). Using our technique, we provide, for the first time, high-resolution, 3D measurements of electromechanical waves, including electromechanical vortices during VF, as well as measurements of APD and contractile changes during pharmacological blockage of potassium ion channels in an isolated intact contracting rabbit heart. The videos are available in the online version of this article. Interactive 3D renderings are also available online (https://cardiacvision.ucsf.edu/videos/3d-optical-mapping).

Sinus rhythm

Figures 6–12 (see also Supporting information, Videos SV3, SV4, SV6, SV10 and SV11) show that we can track the entire strongly deforming 3D ventricular surface during sinus rhythm. Figures 7*B*,*C*, 8 and 12*B* (see also Supporting information, Videos SV3, SV12 and SV13) show that we can subsequently map action potential wavefronts (upstroke) and action potential waves (time-course) propagating across the entire, strongly deforming 3D ventricular surface with non-ratiometric and ratiometric imaging. In particular, with ratiometric imaging, we can

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measure the time-course of the action potential (Figs 1*G*, 8*A*, 10, 11 and 12*D*) and measure its APD (Figs 12*D*,*E* and 13).

Figures 1*F*, 4*E* and 6–12 show the reconstructed 3D ventricular surface together with displacement vectors (red) indicating the motion and deformation of the heart as it contracts in response to the depolarization of the ventricles. The vectors in Figs 4*E*, 6*C*,*E* and 7*A*,*C* indicate displacements of the epicardial surface with respect to its diastolic mechanical configuration just before the onset of the depolarization (at 0 ms), whereas the vectors in Figs 1*F*, 6*D* and 12*D*,*E* indicate

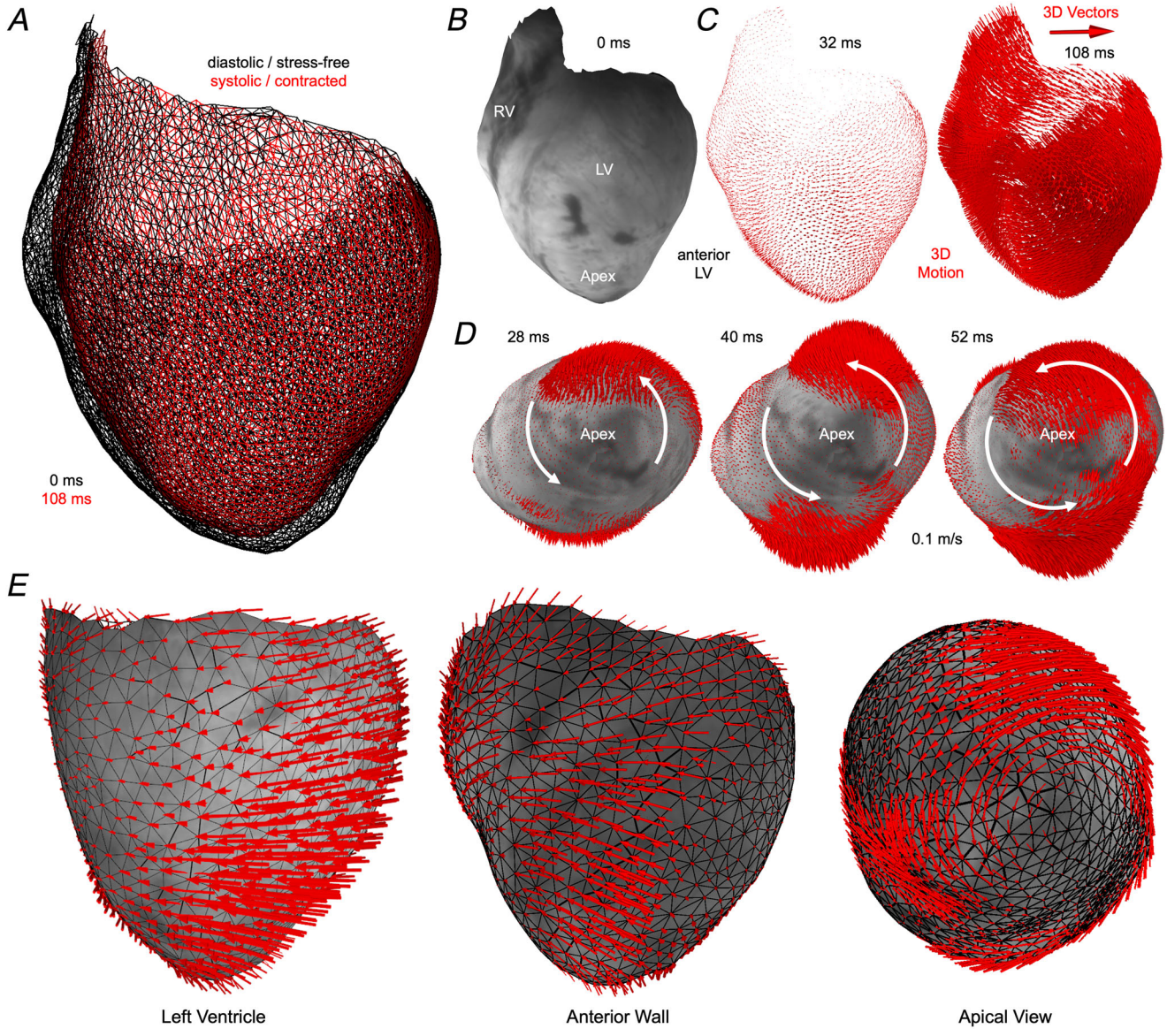


Figure 6. Ventricular wall motion imaged with voltage-sensitive 3D optical mapping during sinus rhythm (see Figs 7 and 12)
Wall motion is tracked in 3D space using video data obtained with 12 calibrated high-speed cameras (see Figs 1 and 4). *A*, wireframe meshes of epicardial surface in diastolic, stress-free (black) and deformed, contracted (red) mechanical configuration. The mesh consists of 3325 vertices or 6349 polygon triangles. Average nearest-neighbour distance: 0.92 ± 0.21 mm (black mesh). *B*, raw video images of left ventricle (LV) projected onto mesh (same heart as in Fig. 6*A* , anterior wall). *C*, 3D displacement vector field (red) indicating motion at 32 and 108 ms with respect to diastolic, stress-free mechanical configuration at 0 ms. *D*, apical view with displacement vectors (red) indicating motion with respect to the previous frame. The fastest contractile motion occurs between 30 and 50 ms, and the ventricles exhibit rotational, torsional motion around the apex (see Supporting information, Videos SV2 and SV12). *E*, additional views of texturized triangular polygon mesh and displacement vectors (red) (anterior wall, posterior wall, apical view), same data as shown in Figs 4 and 9*A*–*C*.

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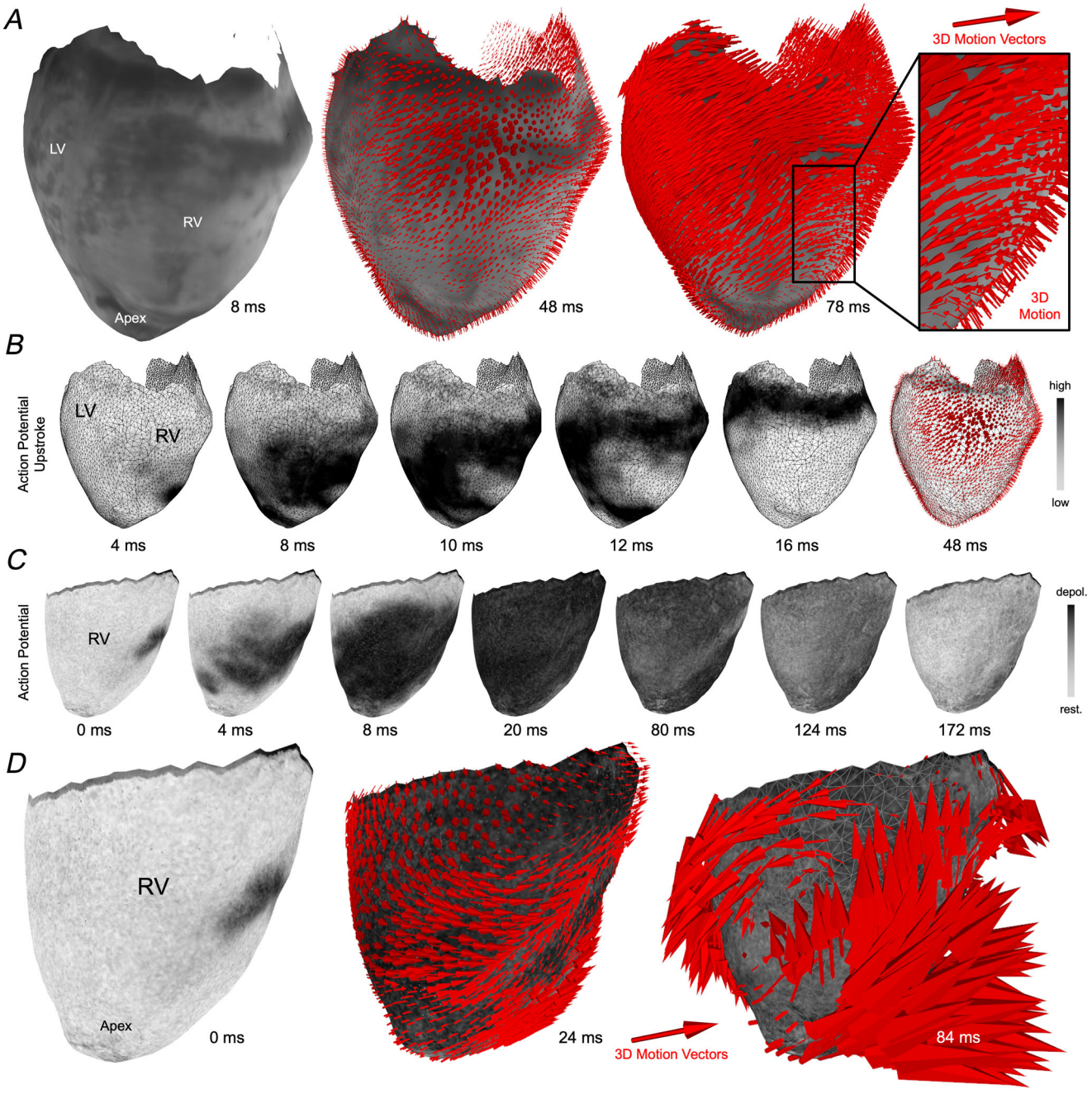


Figure 7. 3D optical mapping of sinus rhythm in beating heart
Sinus rhythm imaged in a contracting rabbit heart with voltage-sensitive multiple-camera optical mapping with 12 cameras (see Figs 1, 6, 8–12; see also Supporting information, Videos SV3, SV12 and SV13). Using 3D motion tracking, we imaged action potential waves and wavefronts across the entire contracting and strongly deforming ventricular surface (LV/RV: left/right ventricle). *A*, raw greyscale video data mapped onto 3D reconstructed ventricular surface together with motion vectors (red) indicating motion and deformation of the ventricular surface with respect to the mechanical configuration at 0 ms (beginning of depolarization). *B*, action potential wave front (dark: upstroke, high negative rate of change in fluorescence intensity) imaged with continuous green excitation propagating across ventricular surface triggering the contraction and deformation of the heart (see Fig. 12*A* and *B*; see also Supporting information, Video SV12). Noticeable contraction sets in ∼30 ms after the beginning of the depolarization of the ventricles (c.f. Fig. 7*C* and *D*). *C*, action potential wave (bright: resting tissue, black: depolarized tissue/plateau phase, grey: repolarizing tissue) measured with pulsed green-blue ratiometric excitation across ventricular surface in strongly contracting heart (see Fig. 8; see also Supporting information, Videos SV3 and SV13). Corresponding optical traces shown in Figs 1*G*, 8*A*, 10 and 12*D*. *D*, close-up with motion vectors (red) as in *A*) (but with different scaling, subsampled).

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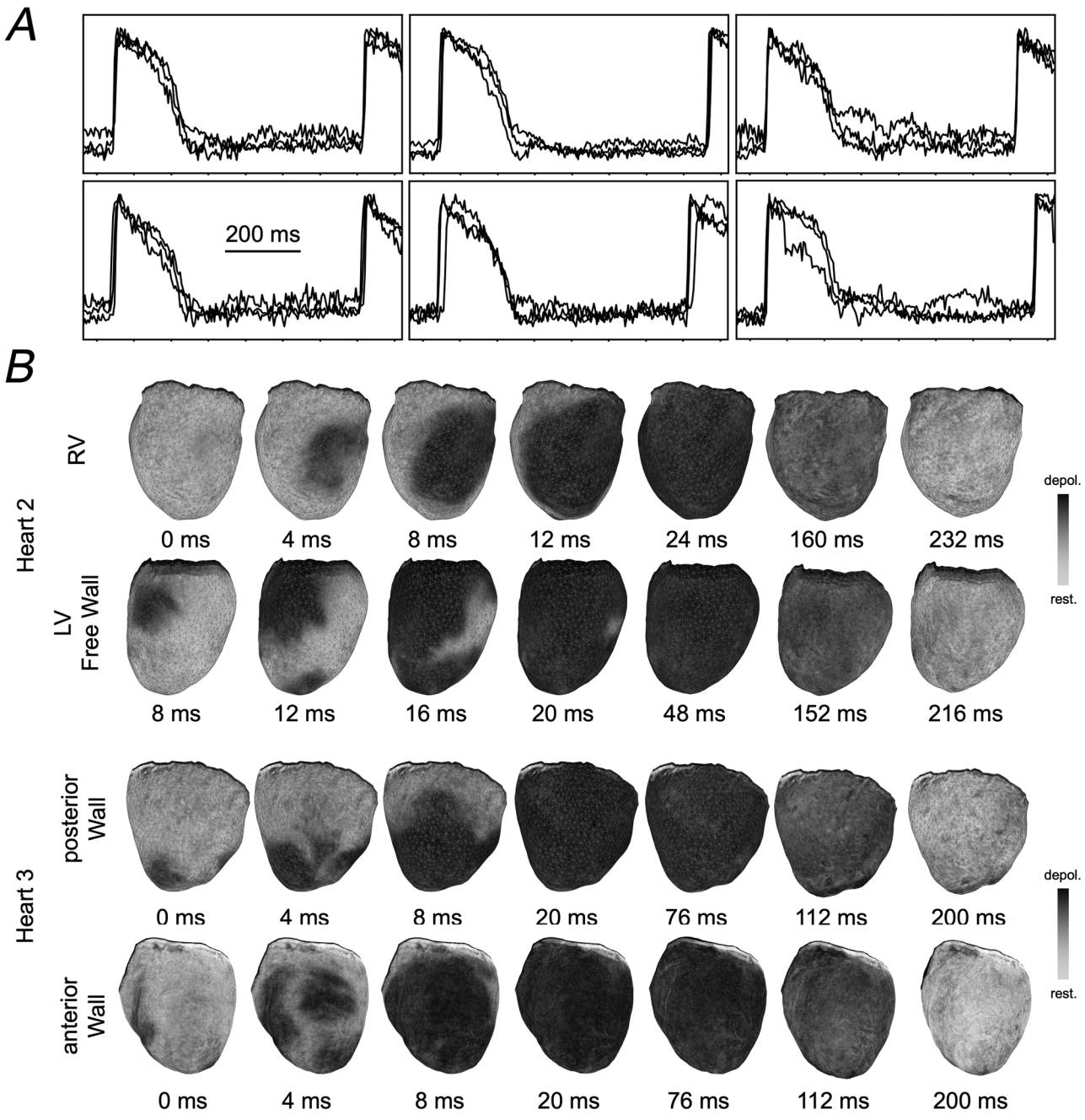


Figure 8. Optical measurement of action potentials across epicardial surface of strongly contracting rabbit hearts
A, optical traces (three in each panel) showing ventricular action potentials during sinus rhythm measured in random locations (out of ∼3000 triangles per heart) across strongly contracting ventricular surface of rabbit hearts ($n = 2$). Each trace was averaged from a 7 $\sim$ 7 pixel region (at centre of each triangle). *B*, optical maps showing action potential wave (black: depolarized tissue) during sinus rhythm in two different hearts (LV/RV: left/right ventricle) (see Supporting information, Videos SV3 and SV13). The data were obtained in ratiometric mode, as also shown in Fig. 7*C* and *D*, and the optical maps were pixel-wise normalized over a 1 s window without spatio-temporal smoothing. No collimator lenses (see Fig. 5). In total, we obtained similar results in $n = 4$ hearts.

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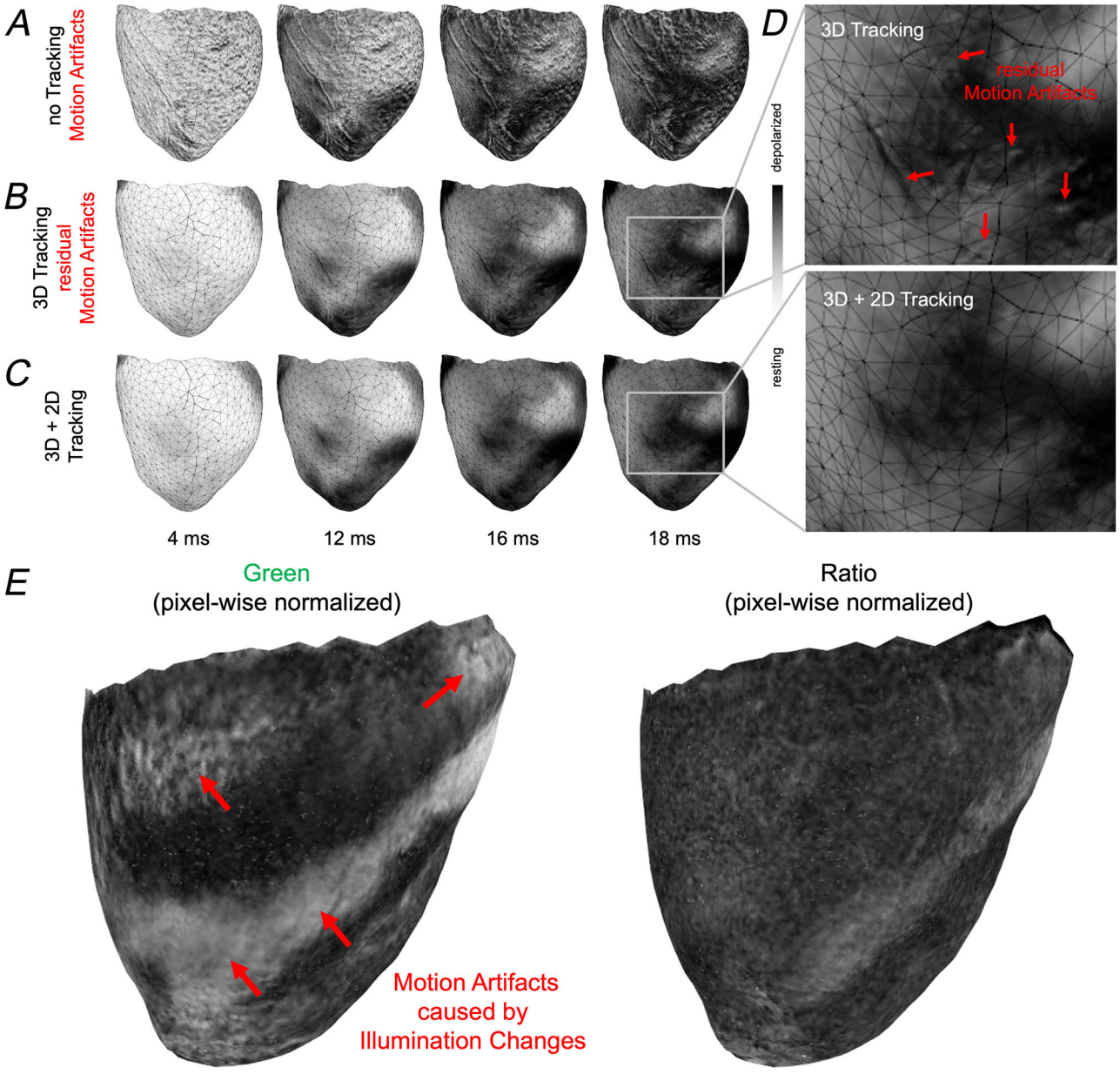


Figure 9. Motion artifact compensation using 3D tracking and ratiometry
Motion artifacts can be significantly reduced using a combination of 3D tracking, 2D tracking and ratiometry. The following (pixel-wise normalized) 3D optical maps show the effect of each step. *A*, no tracking. The optical maps exhibit strong motion artifacts (high-frequency spatial pattern). *B*, only 3D tracking. The maps exhibit residual motion artifacts, because the spatial resolution of the 3D tracking is lower than the resolution of the video texture. *C*, 3D tracking with subsequent 2D tracking (all data in this study were obtained in this way) shows further improvement and inhibition of motion artifacts. *D*, close-ups of motion artifacts. All data in (*A*) to (*D*) were acquired during sinus rhythm with continuous green excitation and post-processed using a sliding-window normalization (100 ms interval). *E*, ratiometric imaging was used to further suppress motion artifacts after tracking. These artifacts arise due to relative motion between LEDs and tissue (see Figs 1*F*,*G*, 5, 10 and 11). Optical maps showing right ventricle during plateau phase of the action potential (dark: depolarized tissue). The corresponding optical traces were obtained using pixel-wise normalization (over 1 s long recording) and are shown in Fig. 10. Left: 3D+2D tracking without ratiometry (green channel only). Right: 3D+2D tracking with ratiometry (green and blue channels). Both optical maps were equally processed in a co-moving frame of reference. Without ratiometry, the optical maps exhibit motion artifacts (bright areas) that stem from illumination changes as the heart moves back and forth or deforms, facing the excitation light sources differently. With ratiometry, the optical maps are more homogeneous. Excitation with collimator lenses (see Fig. 5).

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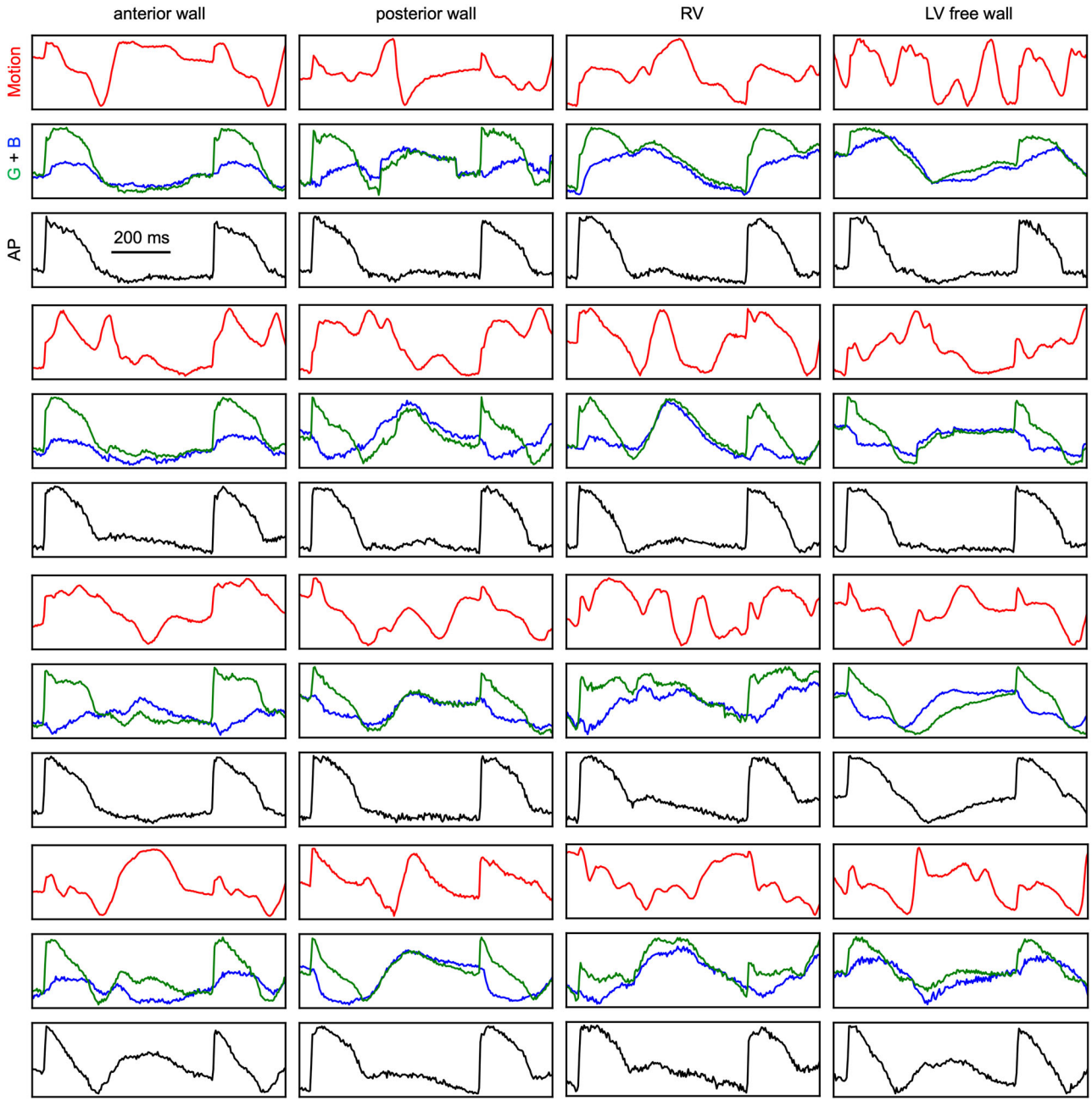


Figure 10. Efficacy of co-moving signal analysis and motion artifact compensation
Optical traces (4 × 4 panels) demonstrating efficacy of combined 3D motion tracking and ratiometry for motion artifact compensation (see Fig. 11). Red: motion artifacts before tracking (processed data with static non-moving mesh). Green–blue: optical traces obtained with green and blue excitation after tracking. Only the green channel exhibits signal. Both channels indicate residual motion artifacts (baseline changes), which are caused by changing illumination as the heart moves. The blue channel mostly tracks the baseline of the green channel. Black: optical traces showing action potentials after ratiometric compensation. Residual artifacts improve without collimators (see Fig. 11) and can be compensated using the baseline correction routine shown in Fig. 14. Each trace was averaged from a 7 × 7 pixel region located at the centre of one of the triangles of the mesh representing the ventricular surface (c.f. Fig. 4*E*) and pixel-wise normalized (dimensionless units). The triangles were selected from random locations on either the anterior wall (first column), the posterior wall (second column), the right ventricular wall (RV, third column) or the free left ventricular wall (LV, fourth column). The data were obtained with collimator lenses mounted onto the LEDs, see Fig. 5.

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instantaneous displacements between subsequent frames. The instantaneous displacements reflect contractile speed (Fig. 12*D*) (absolute values). The data in Fig. 6*D* show that the heart exhibits torsional motion around the apex. It moves upwards as it contracts, whereas its base remains roughly in its vertical position (Fig. 6*A*). The 3D tracking allows us to compensate motion artifacts and measure optical traces across the heart surface in a co-moving frame of reference (Fig. 4*E*; see also Supporting information, Video SV5). Furthermore, we can compute optical maps showing the spatio-temporal electrical activity per pixel (on each triangle) and correlate

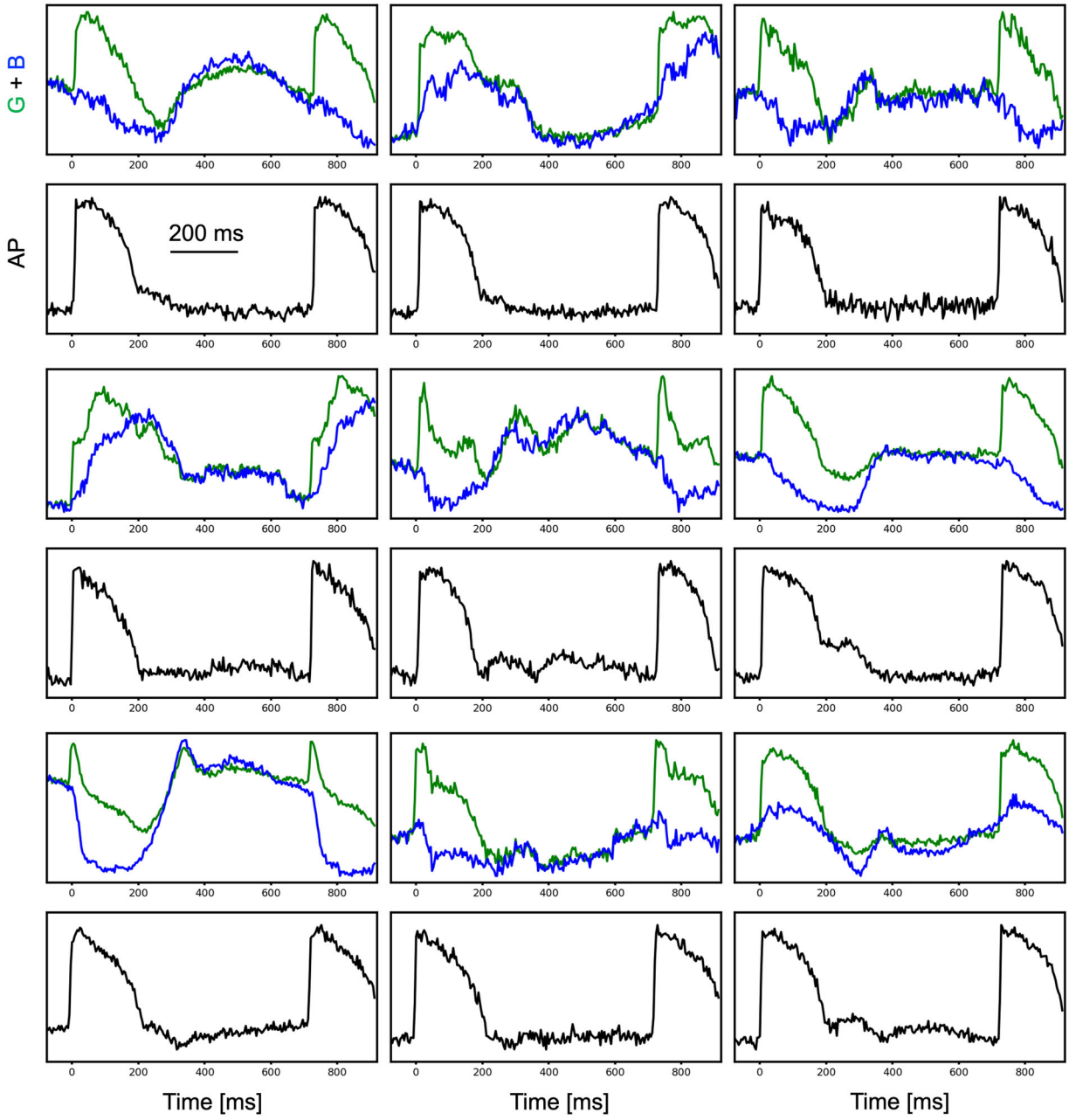


Figure 11. Efficacy of refined motion artifact compensation with ratiometry
Optical traces (3 × 3 panels) demonstrating the efficacy of combined 3D motion tracking and ratiometry for motion artifact compensation (see Fig. 10). Here, the data were obtained without collimator lenses (see Fig. 5), which appears to improve the homogeneity of the light field inside the imaging chamber. All traces were obtained after 3D numerical motion tracking and sampled from a 3 × 3 pixel region at the centre of a random triangle from the surface of the heart shown in Fig. 8*B* (Heart 2) (see also Supporting information, Videos SV3 and SV13). The green and blue traces were aligned vertically by subtracting the absolute difference of their baseline. Otherwise, they were unaltered (no scaling).

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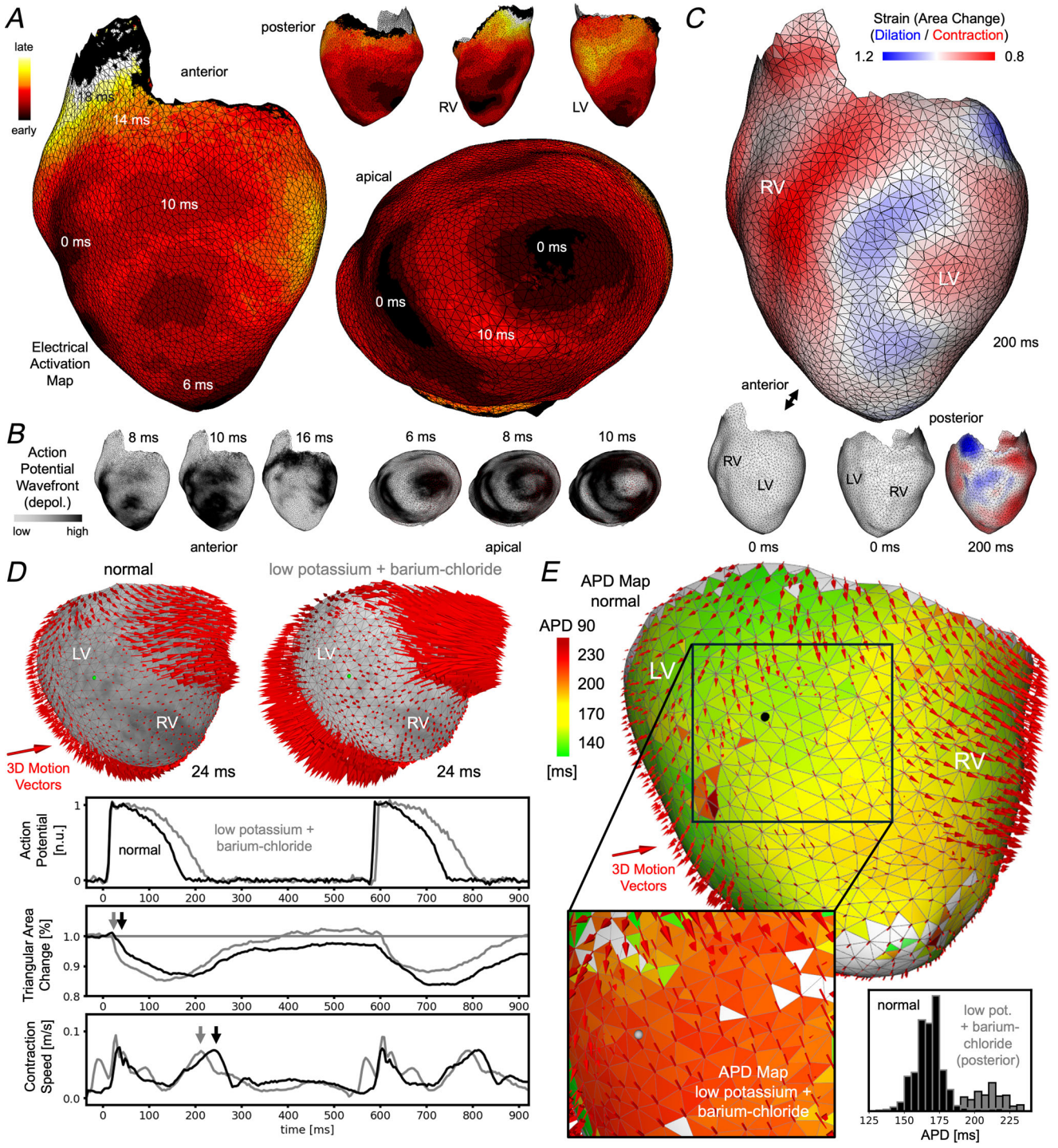


Figure 12. Electrical activation, strain, and action potential duration (APD) mapped across entire ventricular surface of isolated beating rabbit heart during sinus rhythm
A, electrical activation map with local activation times (black: early/0 ms, yellow/white: late/20 ms; anterior, posterior, apical view, etc.) computed from action potential upstroke in (*B*) and Fig. 7*B*. The activation map indicates multiple early-activation or epicardial break-through sites (presumably correlated with the Purkinje system). The entire ventricles get electrically activated within 20 ms. *B*, action potential wavefront (black: upstroke, high negative rate of change of signal; 0 ms: beginning depolarization, continuous green excitation) as seen from anterior and apical walls (c.f. Fig. 7*B*). *C*, strain (measured as local triangular area change) across epicardial surface (blue: dilated, red: contracted, 1.2/0.8 = 20% area increase/decrease, anterior wall). Bottom left: undeformed, stress-free epicardial surface (0 ms: beginning depolarization, anterior wall). Bottom centre

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and right: stress-free and deformed epicardial surface (0 *vs*. 200 ms, posterior perspective) (c.f. Fig. 7). *D*, top: comparison of contraction (red vectors, frame-to-frame displacements) with normal and low potassium barium-chloride Tyrode. Upper: action potential on posterior wall (black dot in *E*) during normal (black) and abnormal sinus rhythm with low-potassium barium-chloride Tyrode (grey) causing prolonged APD. The optical traces were measured at the centre of the same triangle (averaged from 7 ×7 pixels, ratiometric imaging with pulsed green–blue excitation). Centre: strain (local trianglular area change, normalized with respect to initial triangle area) is comparable but rate of contraction is faster/stronger with barium-chloride (arrows). Bottom: contractile speed (magnitude of frame-to-frame displacements) shows that the heart contracts more strongly and relaxes earlier with barium-chloride (arrows). *E*, APD maps computed for normal APD across entire ventricular surface and for prolonged APD on posterior wall during sinus rhythm (APD-90 at 10% height of the action potential); for additional data, see Fig. 13 for additional data. Histogram showing peaks at 170 ms (SD = 5 ms) with normal Tyrode and 210 ms (SD = 5 ms) with barium-chloride (only posterior wall). Traces in (*D*) sampled from black and grey dots (same location, same triangle). Heart in (*D*) and (*E*) different than in (*A*) to (*C*).

the electrical activity with the mechanical contraction of the heart. The action potential wavefront (black) shown in Figs 7*B* and 12*B* (measured with continuous green excitation) corresponds to the action potential upstroke, which was computed in the co-moving frame as the difference between pixel intensities in subsequent frames (pixel-wise normalized temporal derivative of the optical maps). The action potential wave originates in several locations activating the entire ventricular muscle within less than 20 ms; see corresponding electrical activation map (Fig. 12*A*; see also Supporting information, Video SV12). The ventricles begin to contract and deform shortly thereafter (Figs 6*C*,*D* and 7*B*). The epicardial surface experiences mostly contractile strain with a local area decrease of ∼10–20% (triangular area change) and tensile strain in some areas with a local area increase of ∼5–15% (Fig. 12*C*). Noticeable contraction sets in only about 30 ms after the onset of the depolarization of the ventricles. The ventricles exhibit maximal contraction velocities of ∼0.1 m s−1 around 40–50 ms after the onset of the depolarization (Fig. 6*D*). These high contraction speeds then lead to the large deformations depicted by the wireframe mesh in Fig. 6*A* and the corresponding displacements shown in Fig. 6*C*. Our data show that the depolarization phase of the action potential and the onset of contraction are clearly separated in time. By contrast, the action potential waves (black) shown in Figs 7*C* and 8 (see also Supporting information, Videos SV3 and SV13) (measured with pulsed green-blue excitation) correspond to the time-course of the action potential (pixel-wise normalized optical maps), not just its upstroke. Correspondingly, the depicted wave covers a larger timespan (∼200 ms instead of 20 ms) and the depicted displacement vectors indicate the overall displacement of the ventricular surface during the plateau phase of the action potential. Figures 1*G*, 8*A*, 10, 11, 12*D* and 13 show measurements of the time-course of the action potential for various locations (from centre of a triangle) and Figs 12*E* and 13 show APD measurements across the entire ventricular surface (per triangle). With low-potassium barium-chloride Tyrode, the action potential becomes prolonged (APD increases) and the contraction shortens compared to with normal Tyrode (Fig. 12*D*, lower, arrows). We observe a positive and a negative electromechanical window with normal and low-potassium barium-chloride Tyrode solutions, respectively. A negative electromechanical window is associated with the action potential repolarizing before the heart muscle relaxes (Bekke et al., 2015).

Our data demonstrate that we can measure both the electrical and mechanical dynamics of the heartbeat simultaneously at high spatial and temporal resolutions. The resolution of the electrical measurements, except for the APD measurements, is higher than the mechanical measurements. We measured electrical activation and optical traces per pixel (superimposed data from multiple cameras) and APD and strain per triangle. In the example in Figs. 6 and 7*A*, both the wireframe mesh and the vectorfield consist of 3325 vertices or displacement vectors, with the vertices beinforming 6349 polygon triangles. The average distance between two neighbouring vertices is 0.68 ± 0.30 mm and the average area of a triangle is ∼0.34 ± 0.11 mm^2 (at 0 ms). The entire ventricular surface is resolved by 668,793 pixels, where each triangle comprises about 50–150 pixels (Fig. 4*D*). The spatial resolution of the texture map is so high that it is possible to identify individual small vessels and other anatomical features across the ventricular surface (Figs. 6*B* and 7*A*). Per pixel, the spatial resolution is about 120 μm. The horizontal (short-axis) diameters of the heart are 2.69 cm and 3.02 cm (along the shorter and longer axes of the elliptical shape of the heart).

Electromechanical waves during paced rhythms

To investigate the relationship between electrical activation and mechanical deformation, we imaged focal action potential waves propagating across the ventricles away from a stimulation site (Figs 15-17 ; see also Supporting information, Videos SV14–SV16). The pacing pulses were applied using a thin, flexible stimulation electrode, which was positioned on the ventricular epicardial surface (Fig. 3). Using our multiview reconstruction methodology, we were able to track the

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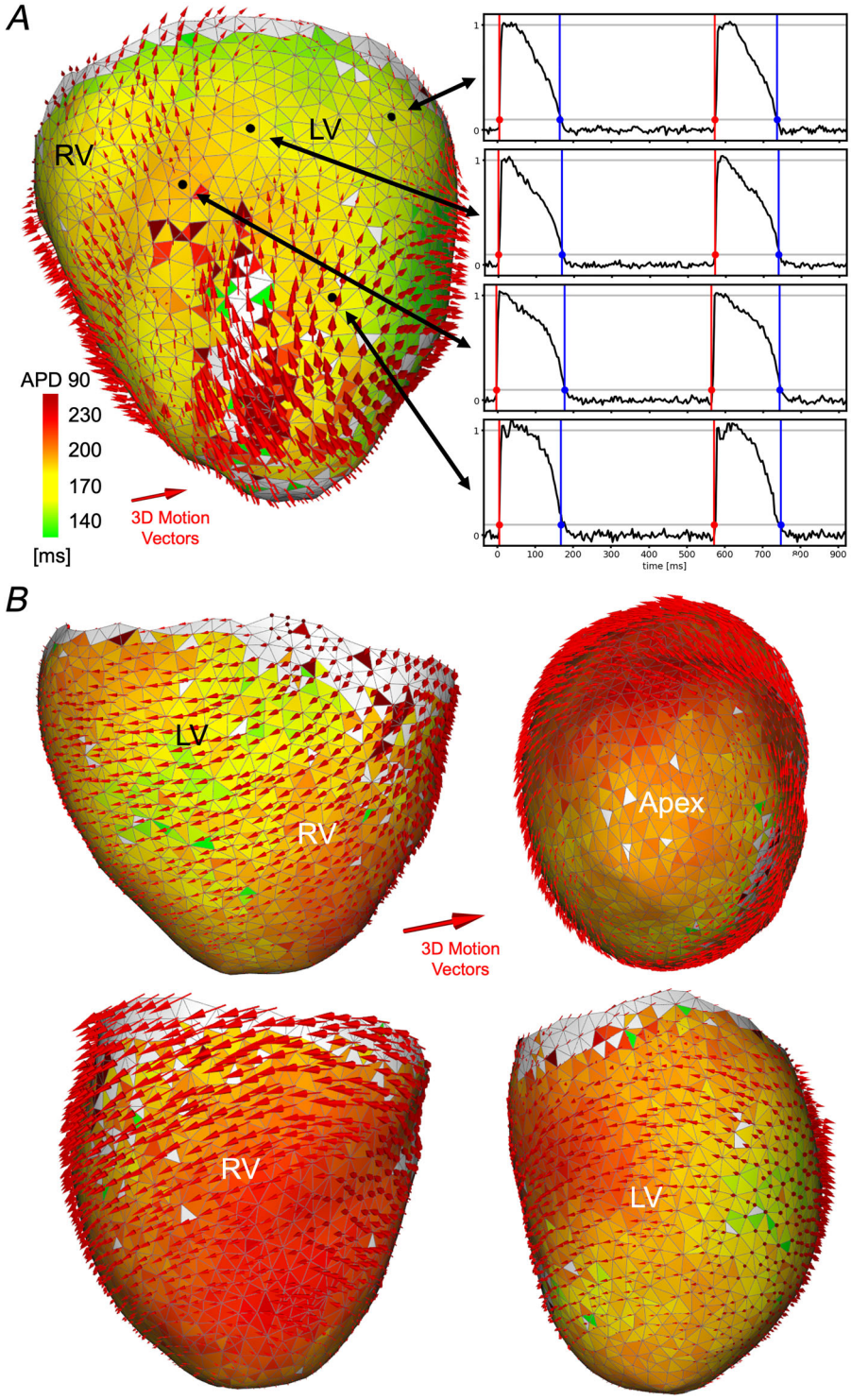


Figure 13. Optical action potential duration (APD) measurements across strongly contracting ventricular surface of rabbit heart
Panoramic APD maps computed from voltage-sensitive optical mapping data showing APDs across entire strongly contracting ventricular surface of rabbit hearts (n = 2) during sinus rhythm. *A*, APD computation at 10% height of the action potential (grey line, APD 90) using upstrokes (red) and repolarization times (blue) shown for four representative sites. Values in APD map (anterior wall) averaged from two subsequent action potentials. An APD measurement was rejected/excluded (white triangles) either when the first and second APDs were significantly different or if there was an inconsistent number of upstrokes or downstrokes. Traces were pre-processed as shown in Fig. 14. Same heart as in Figs 7*C*,*D* and 12*D*,*E*. *B*, additional example with slightly longer APDs (mean = 192 ms, SD = 35 ms) measured in 3380 locations (triangles) and APD gradient from anterior to posterior wall. Each APD

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measurement stems from a single action potential per trace. An APD measurement was rejected/excluded (white triangles) if multiple or no upstrokes were detected (here 462 triangles or ∼14% of the data). Same heart as in the Supporting information (Videos SV3 and SV13). LV/RV: left/right ventricle. Red displacement vectors indicate movements from one frame to the next (arbitrary scaling of magnitude).

tissue underneath the electrode as it is always visible in at least two or three of the other camera images (Fig. 3*A* and *B*).

Figure 15*A* and *B* shows a focal action potential wavefront (black) propagating across the ventricles after the application of a pacing pulse during continuous pacing at 4.5 Hz. The electrical wavefront corresponds to the upstroke or depolarization phase of the action potential (non-ratiometric imaging with continuous green excitation) and therefore to the moment when the tissue becomes electrically activated. The electrical activation wavefront propagates away from the stimulation site towards the opposite side of the heart. We computed an electrical activation map from this data (Figs 16*E* and 17*A*; see also Supporting information, Video SV17). The electrical activation map exhibits an elliptical pattern around the pacing location, reflecting the anisotropic conduction of the action potential wave. At the same time, the heart contracts and deforms in response to the electrical activation. Displacement vectors (red) indicate

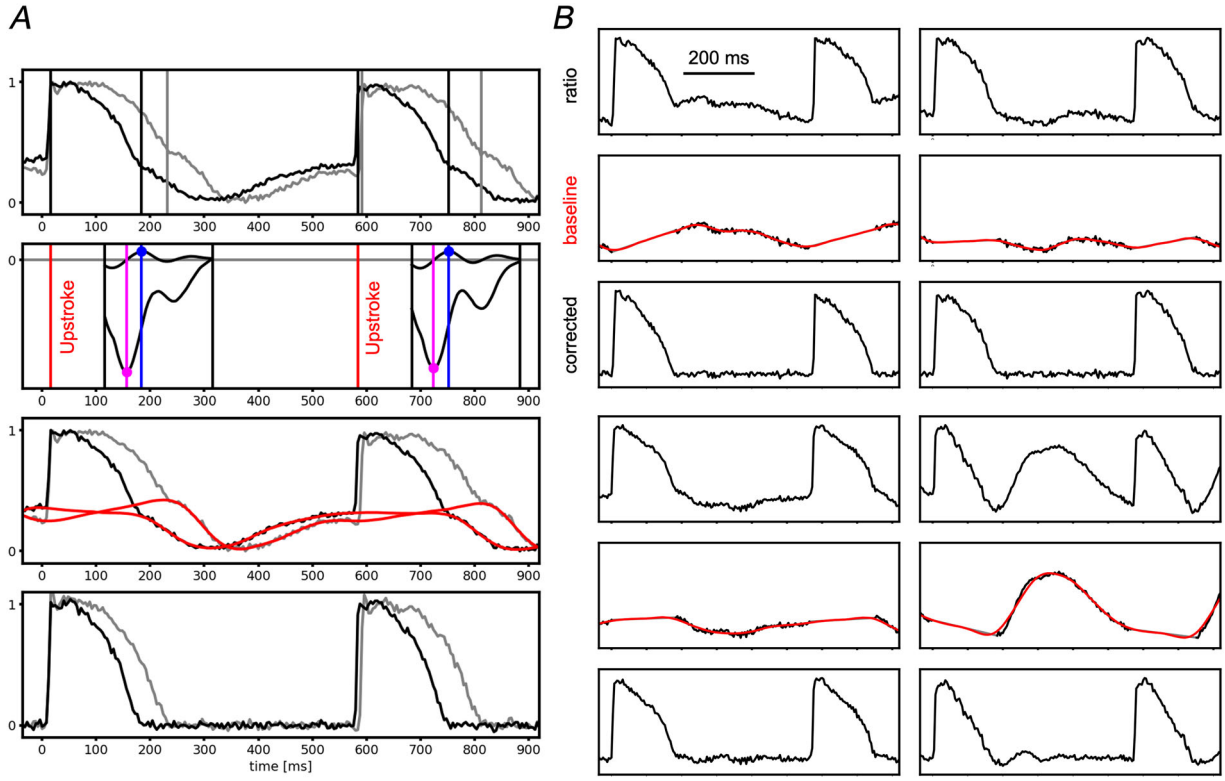


Figure 14. Baseline correction for action potential duration (APD) measurements
Custom baseline correction method for removal of residual motion artifacts, which present as baseline modulations, after motion tracking and ratiometric compensation. All data were obtained with collimator lenses mounted in front of the LEDs (see Fig. 5). *A*, baseline detection and removal. Top: optical traces with normal (black) and low-potassium (grey) Tyrode (see Fig. 12*D*). Bars: upstrokes and repolarization times. Second panel: first and second (temporal) derivatives of the optical trace (both black) with normal Tyrode within a defined window (between 100 and 300 ms after each upstroke). Upstroke (red bar) detected as maximum in first derivative. Steepest downslope of repolarization phase (pink bar) and completion of repolarization (blue bar) detected as extremal points of first and second derivatives, respectively. Accordingly, repolarization times were detected as a sudden change from the steepest part of the repolarization phase to baseline. The trace and derivatives were smoothed using Savitzky–Golay filtering prior to each processing step. Third panel: using the upstroke and repolarization times, the action potential was removed and missing values were linearly interpolated. The resulting trace was smoothed using Savitzky–Golay filtering (red). Bottom: subtraction of the fitted baseline yields the corrected optical traces. *B*, representative examples of baseline corrections (2 × 2 panels), which yield consistent repolarization times, flat diastolic intervals for most optical traces, and uniform APD measurements across the ventricular surface (see Figs 12*E* and 13). The baseline correction was only used for APD measurements.

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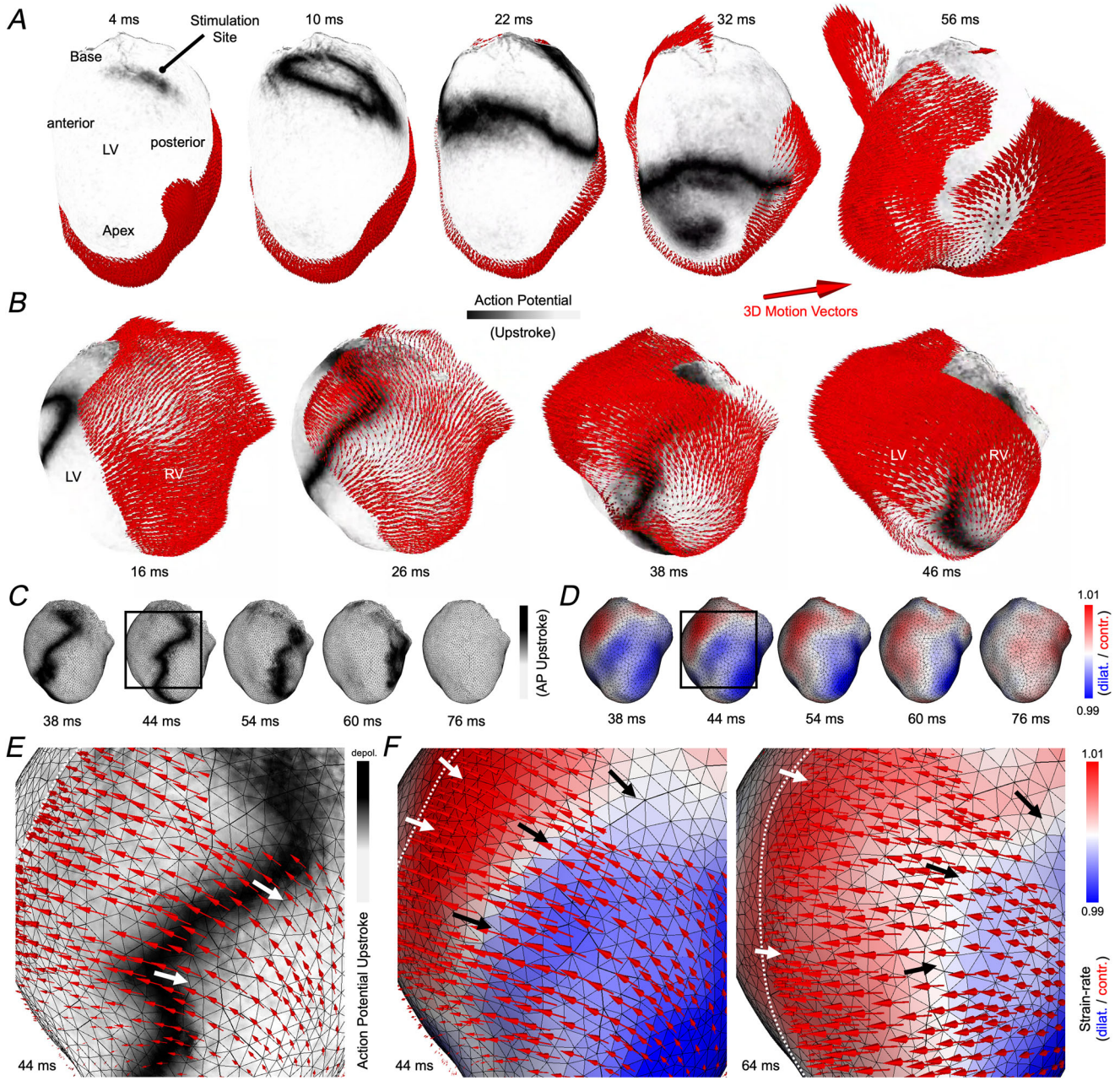


Figure 15. Electromechanical wave during pacing in contracting isolated rabbit heart (see Supporting information, Videos SV14 and SV15)
A, action potential (AP) wavefront propagating away from stimulation site (4.5 Hz pacing frequency) across the deforming ventricular surface, see Fig. 16. Black: depolarizing tissue/action potential upstroke (temporal derivative of optical trace, continuous green excitation). Red: 3D displacement vectors indicating motion with respect to previous frame acquired 4 ms earlier (vectors pointing into the heart not visible). Frontal view showing focal wave propagating across left ventricular (LV) free wall away from stimulation site. The ventricles get electrically activated within about 50 ms. *B*, side view showing action potential wave propagating away from stimulation site across posterior wall and right ventricle (RV) towards opposite side of the heart. The tissue starts moving towards the approaching wave and contracts noticeably when the AP wavefront has propagated across and activated the whole muscle. *C*, electrical activation wavefront (black: action potential upstroke) propagating across posterior wall (c.f. *A*). *D*, mechanical activation wavefront consisting of a transition from dilating (blue) to contracting (red) strain-rates measured as the rate of triangular area change (±30% per 100 ms) of each mesh triangle. The mechanical wave follows the electrical wave with a short delay of ∼5 ms (see Fig. 16*G*). The phenomenon occurs on both the

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posterior and anterior walls (see Figs 16 and 17*B*,*C*). *E*, close-up of electrical wave (white arrows) together with tissue displacement vectors (red, indicate motion with respect to the tissue configuration 4 ms earlier). The tissue is being pulled towards the approaching electrical wave. *F*, close-up of mechanical wave (black vectors) with dilating tissue (blue) in front of the electrical wave and contracting tissue (red) shortly after the electrical activation behind the wavefront (see Supporting information, Video SV16). The vectors undergo a rapid change in direction within the contracting region (white dotted line). The data were obtained with continuous green illumination at 500 frames s^{-1}. The electromechanical wave consists of two closely coupled electrical and mechanical activation waves (see Fig. 16).

the motion of the epicardial surface (with respect to the mechanical configuration in a previous frame, acquired 4 ms earlier) (see Supporting information, Video SV15). Initially, the heart wall contracts close to where the pacing pulse is applied, but the contractile motion sets in following the electrical activation. Shortly thereafter (30–40 ms) the wall on the opposite side moves towards the origin of the wave. Note that only vectors pointing outwards, away from the epicardial surface are visible, whereas vectors pointing beneath the surface are hidden behind the texturized surface. The data demontrate a strong correlation between the ventricle's electrical activation and motion. Here, motion is represented as kinematic data or as a displacement vector field.

Figure 15*C*–*F* shows that the focal action potential wave causes the ventricles to deform in a very characteristic way: as the muscle contracts close to the pacing location, tissue further away is being pulled toward the contracting part of the heart. Subsequently, the tissue that is being pulled experiences dilatation or tensile stretch. Moreover, as the electrical wavefront spreads across the ventricles, the contraction closely follows the electrical wave (Fig. 16; see also Supporting information, Videos SV14 and SV16). Plotting the epicardial strain-rate (measured as the rate of triangular area change), we see that the electrical wavefront is immediately followed by a mechanical wavefront that is defined as the transition from dilating (blue) to contracting (red) strain-rates (Fig. 16*C* and *D*). As the tissue is pulled towards the approaching electrical wave (red vectors) it dilates (blue) and then suddenly contracts (red) as the electrical wave passes through. This mechanical activation front follows the electrical activation front with a short delay of $\sim 5.0 \pm 2.4$ ms as it propagates across the posterior and anterior ventricular walls (Fig. 16*C*, *D* and *G*). Based on this observation, we computed the mechanical activation times as the transition from positive or dilating to negative or contracting strain-rates (Fig. 16*D*) and subsequently computed a mechanical activation map (Fig. 16*F*), which exhibits similarities to the corresponding electrical activation map (see Supporting information, Video SV17). The delays between electrical and mechanical activation times were calculated for the points (red) shown in Fig. 16*G*. Additionally, we computed the propagation speeds of the electrical and mechanical wavefronts across the posterior wall along the path shown in Fig. 16*H* (between one to the next as well as to the second-next point), and the speeds are comparable (electrical: 0.55 ± 0.26 m s^{-1}; mechanical: 0.49 ± 0.27 m s^{-1}).

It is important to note that, although the electrical and mechanical activation times correlate well across a wide area of the ventricles, they do not correlate well everywhere, particularly in the vicinity to the pacing electrode (see Supporting information, Video SV17).

Ventricular fibrillation

Hearts contract only minimally during VF. However, the residual contractile motion during VF is yet strong enough to produce substantial motion artifacts, as shown in Fig. 18*B*. In this study, we discuss two episodes of VF: one with very little to almost no motion (VF1) (Figs 18 and 19*A*; see also Supporting information, Video SV18) and one with moderate contractile motion (VF2) (Fig. 19*B*,*C*; see also Supporting information, Video SV19). The different contractile strengths between the two VF episodes can be associated with a much shorter APD and accelerated fibrillatory dynamics (20 Hz) which leads to very minimal motion in episode VF1, *vs*. a longer APD with slower fibrillatory dynamics (10 Hz) resulting in moderate contractile motion in episode VF2 (Fig. 18*E* and *F*). While episode VF1 is an interesting example to discuss motion artifacts, episode VF2 is interesting in terms of its ventricular mechanics. Both episodes were recorded in the same heart.

The VF1 episode in Fig. 18 shows that, even with very little to almost no motion, it is necessary to track and compensate motion in optical mapping studies because, otherwise, motion artifacts become so strong that they prohibit further analysis of the data. Figure 18*B* shows pixel-wise normalized optical maps, which were not tracked and subsequently not analysed in a co-moving frame. Motion artifacts manifest as the high-frequency dark-bright pattern, which superimposes the action potential waves; see fig. 3*C* in Christoph et al. (2018) and figs 3, 5*B* and 7*B*,*D* in Christoph and Luther (2018). By contrast, Figure 18*C* shows pixel-wise normalized optical maps computed using co-moving signal analysis. They do not exhibit any significant motion artifacts, but instead multiple reentrant vortex-like action potential wave patterns. Note that, in Figs 18 and 19, we display the action potential wave and not the action potential wave-

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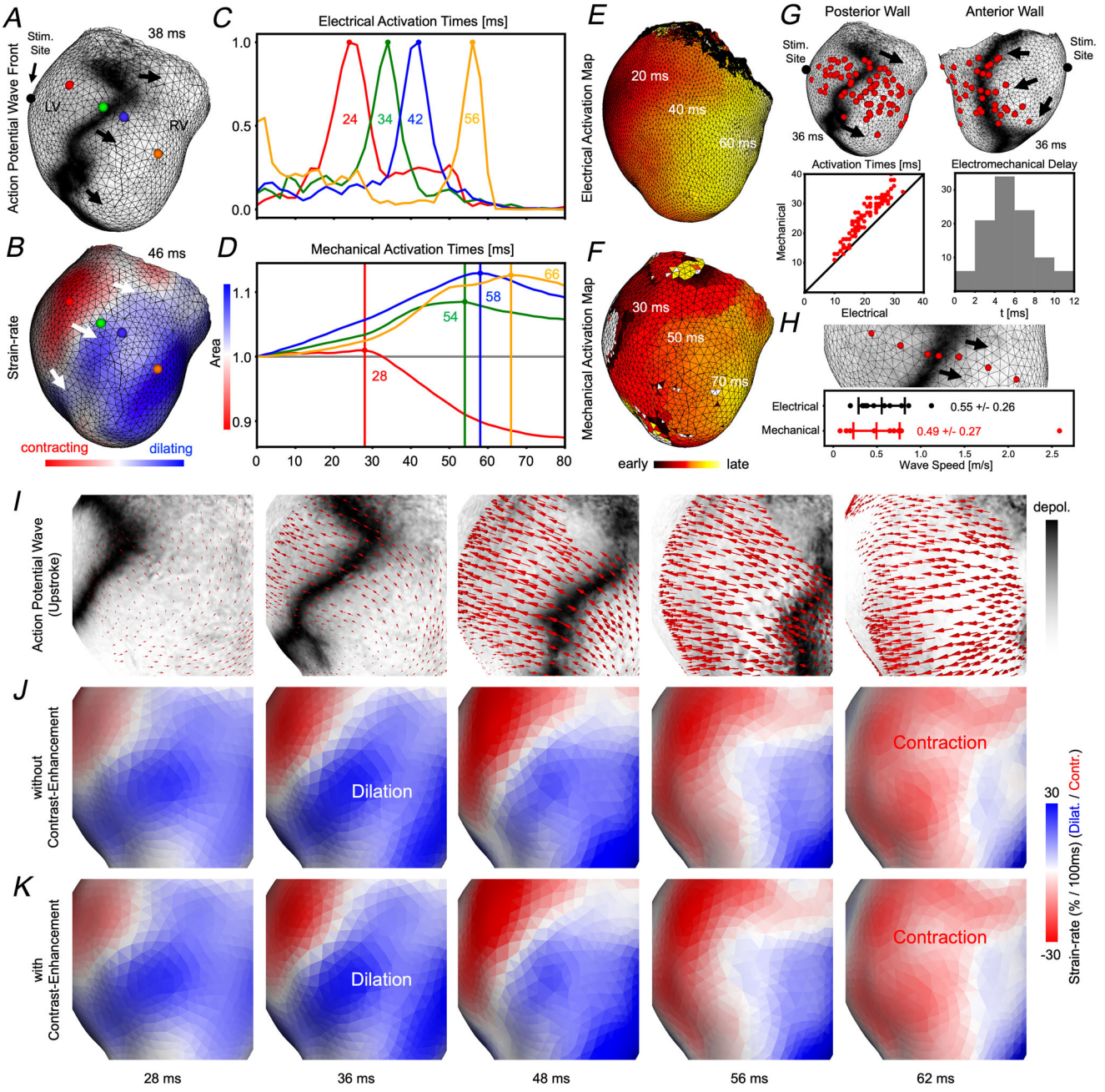


Figure 16. Electrical and mechanical activation during ventricular pacing in intact isolated rabbit heart
A, action potential wavefront (black) propagating across posterior wall activating subsequent measurement points (red, green, blue, orange). *B*, strain-rate measured as rate of triangular area change (blue: increasing, red: decreasing) across 3D ventricular epicardial surface following electrical wave front. *C*, electrical activation times (ms) computed from temporal derivative of voltage-sensitive optical signals highlighting depolarization phase or upstroke of the action potential. *D*, mechanical activation times measured as the transition from positive to negative strain-rates (dilating to contracting, slope of strain or area change) calculated from triangular area change. The triangular area is normalized to 1.0 with respect to the area at the time of the pacing pulse at 0 ms. The mechanical activation times are delayed with respect to the electrical activation times (4, 20, 16 and 10 ms). *E*, electrical activation map computed from activation times. *F*, Mechanical activation map computed from mechanical activation times. *G*, mechanical over electrical activation times measured from various points scattered across posterior and anterior walls, and distribution of electromechanical delays between electrical and mechanical activation times (mean = 5.0 ± 2.4 ms). *H*, speed of electrical and mechanical activation fronts measured along linear path (red dots). Electrical (0.55 ± 0.26 m s^{-1}, CV) and mechanical (0.49 ± 0.27 m s^{-1}) wave speeds. *I*, action

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potential wavefront (black, red vectors: motion). *J* and *K*, strain-rate patterns (red: contracting, blue: dilating) computed when tracking was performed without and with contrast-enhancement (on the contrast-enhanced video data instead of the original video data), as described in Christoph and Luther (2018). Tracking with or without contrast-enhancement yields almost identical strain-rate patterns, which indicates that the tracking is not influenced by the wavefront.

front as in Figs 7*B*, 12*B* and 15. The dark area corresponds to depolarized tissue, the bright area to repolarized tissue (normalized units [0,1]), as in Fig. 7*C*. In both Figs 18*B*,*C* and 19*B*, we amplified intensity fluctuations numerically using pixel-wise normalization with a sliding window (with a length of 50 frames or 100 ms). The data were generated with continuous green excitation without ratiometric motion artifact compensation. Therefore, changes in the excitation were not compensated and are accordingly visible as slight baseline changes (see Supporting information, Videos SV18 and SV19). In episode VF1, the APD is extremely short in the order of

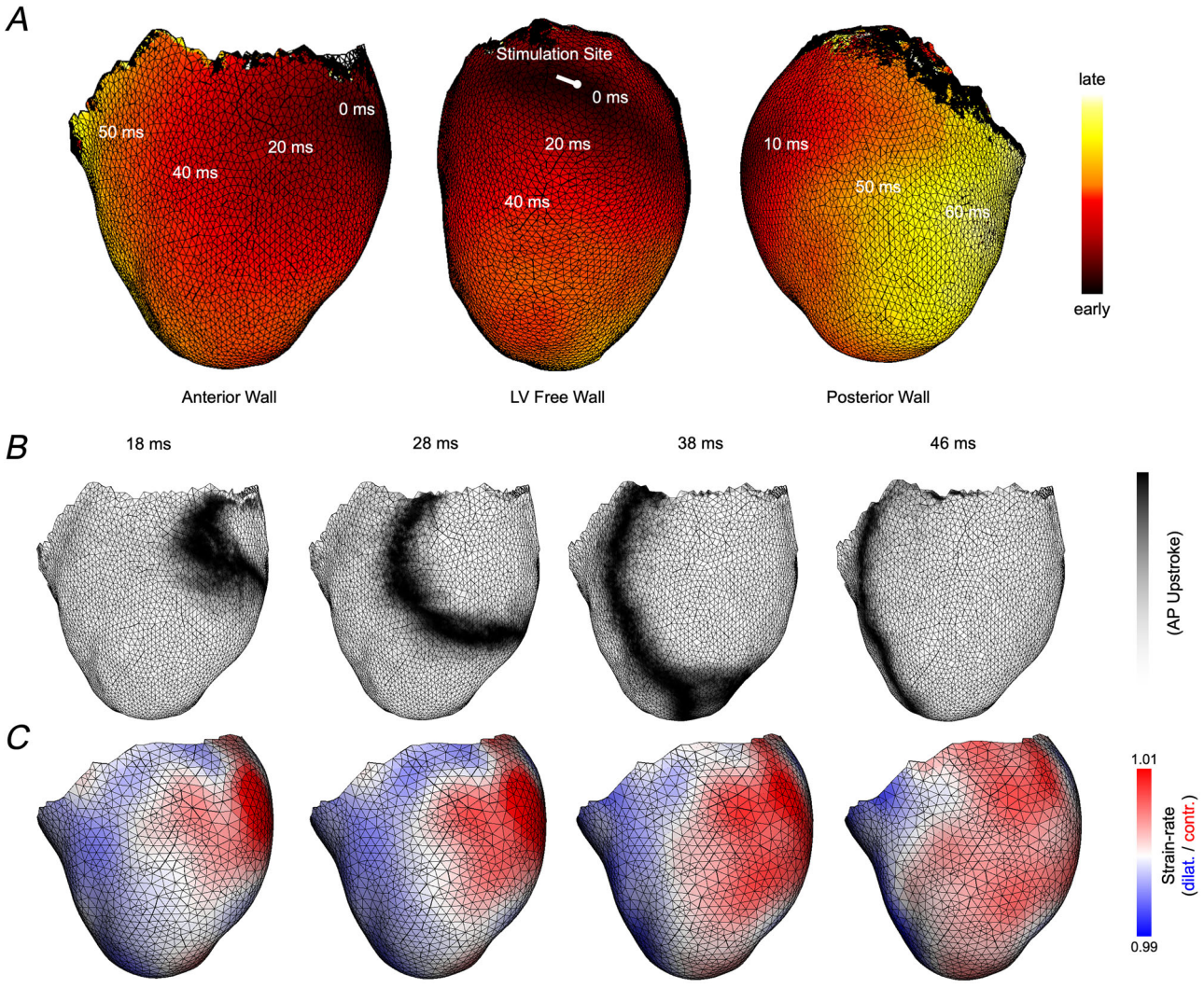


Figure 17. Electrical and mechanical activation during ventricular pacing in intact isolated rabbit heart
A, panoramic electrical activation map measured across strongly deforming ventricles using voltage-sensitive optical mapping. The ventricles become electrically activated within ∼60 ms. The elliptical shape of the electrical conduction and activation on the free left ventricular (LV) wall is associated with the muscle fibre anisotropy of the heart. *B*, action potential wavefront (black: upstroke) propagating across deforming posterior ventricular wall (opposite side of the heart shown in Fig. 16. *C*, mechanical activation wavefront consisting of a transition from dilating (blue) to contracting (red) strain-rates (measured as the rate of triangular area change of each mesh triangle), see Fig. 16*B*,*D*. The electrical and mechanical activation are correlated and suggest that both activation wavefronts constitute an electromechanical wave. However, the data is poorly correlated close to the stimulation site (see Fig. 16*E*,*F*; see also Supporting information, Video SV14).

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30–40 ms and the wavelength is in the order of 1–3 mm. The tissue only moves on average by ∼0.2–0.3 mm (Fig. 18*F*). The fact that the strong motion artifacts in Fig. 18*A* occur at all with this little motion highlights the necessity to track the tissue's motion to compensate motion artifacts. Figure 18*D* shows optical traces before and after motion tracking, with motion artifacts present before tracking and absent after tracking. The top trace shows a measurement without tracking obtained from a single pixel of one of the cameras (close to the red dot shown in Figure 18*A*). The time-series shows an unrecognizable, distorted sequence of action potential waves superimposed by strong intensity modulations, which are caused by the back and forth movement of the tissue (the darker nearby vessel moves across the pixel). The trace below shows the corresponding measurement obtained in a co-moving frame (red dot in Figure 18*A*), again from a single pixel of a single camera, but after 3D and 2D tracking. As a result of the tracking and co-moving signal analysis, the motion artifacts are inhibited and the trace shows instead a sequence of action potentials. The bottom trace shows the averaged signal obtained with multiple cameras. Although ratiometry can effectively reduce measurement artifacts in general, it is not necessarily required during VF (Kappadan et al., 2020). We computed phase maps from the motion-stabilized traces (Fig. 19*A*). The phase maps emphasize the vortex-like structure and topological organization of the action potential wave pattern, and indicate the rotational centres of the electrical vortex waves as phase singular points (PS, white dots). In this example, there are several rotating vortices including a figure-of-eight pattern in the anterior left ventricle and a persistent rotating vortex close to the apex. The phase singularities allow us to automatically track electrical vortex waves across the moving 3D ventricular surface.

With the larger contractions in episode VF2, it is possible to observe reentrant action potential vortex waves inducing vortex-like 3D deformation patterns on the epicardial ventricular surface (Fig. 19*B*, also

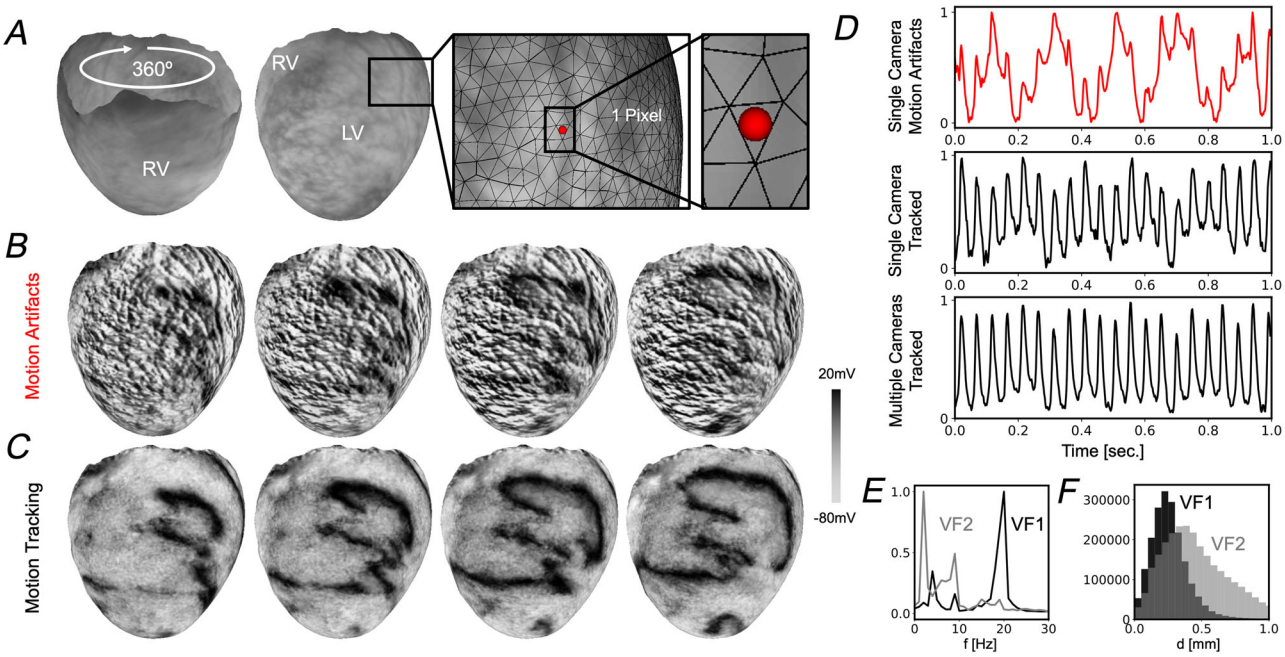


Figure 18. Ventricular fibrillation (VF) imaged across the entire surface of a moving isolated rabbit heart using voltage-sensitive panoramic optical mapping with 12 cameras (see Supporting information, Video SV18)

A, reconstructed 3D ventricular surface and location (red dot) in which optical traces in (*D*) were measured. *B*, optical maps (3D) before motion tracking containing motion artifacts. *C*, tracked and motion-stabilized optical maps (3D) without motion artifacts showing action potential vortex waves on left ventricular (LV) surface (black: depolarized tissue, light grey: repolarized tissue, pixel-wise normalized video, 30–50 frames sliding window as in *B*). *D*, optical traces before (top) and after motion tracking with a single (centre) and multiple (bottom) cameras (measured in one pixel, location in centre of triangle shown in (*A*). *E*, frequency spectra (cumulative, 1 s long video, computed from all motion-compensated optical traces) of fast VF1 episode (black, dominant peak at 20 Hz) shown in (*A*) to (*D*) and (*F*) and 'slow' VF2 episode (light grey, dominant peak at 10 Hz) shown in Fig. 19 (secondary peaks caused by residual motion artifacts). *F*, episode VF1 (black) exhibits minimal motion, while episode VF2 (light grey) exhibits larger contractile motion: distribution of displacement magnitudes *d* [mm] with respect to first frame (1 s long video). Motion artifacts in B are substantial even though the motion is less than 0.5 mm.

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Supporting information, Video SV19). Figure 19*B* shows three examples of spiral wave-like action potential waves (black: depolarized tissue, light grey: repolarized tissue, normalized units [0,1]): a figure-of-eight pattern and two clockwise spiral waves in the left and right ventricles. In this slower episode, the action potential waves induce noticeable contractions, which lead to macroscopic deformations of the ventricles in response to the electrical activation. The dominant VF frequency is ∼10 Hz (Fig. 18*E*). At these slower activation frequencies, the contractions are stronger than at the higher frequencies in episode VF1 in Fig. 18, which is an observation that was also made in Christoph et al. (2018). The tissue moves on average by ∼0.5 mm (Fig. 18*F*). The larger deformations are associated with a spatio-temporal displacement vectorfield (red). Note that the displacement vectors are scaled differently than in Figs 6–16. Here, they display only the direction of the movements of the vertices and not the magnitude of these movements. We chose this visualization to highlight vortices or curls in the displacement vectorfields, which emerge close to electrical phase singularities (PS). Note that displacement vectors pointing into the heart muscle are not visible. Figure 19*C* shows the corresponding phase maps and the mechanical curls in the vicinity of electrical PS. Our data demonstrate a strong correlation between action potential waves and tissue deformation during VF, and hint at topological defects in the contractile motion and deformation dynamics of the heart wall which co-localize with electrical PS.

Discussion

In the present study, we have demonstrated that it is possible to image action potential waves at high resolutions across the entire deforming ventricular surface of contracting isolated hearts using ratiometric voltage-sensitive multiple-camera optical mapping and

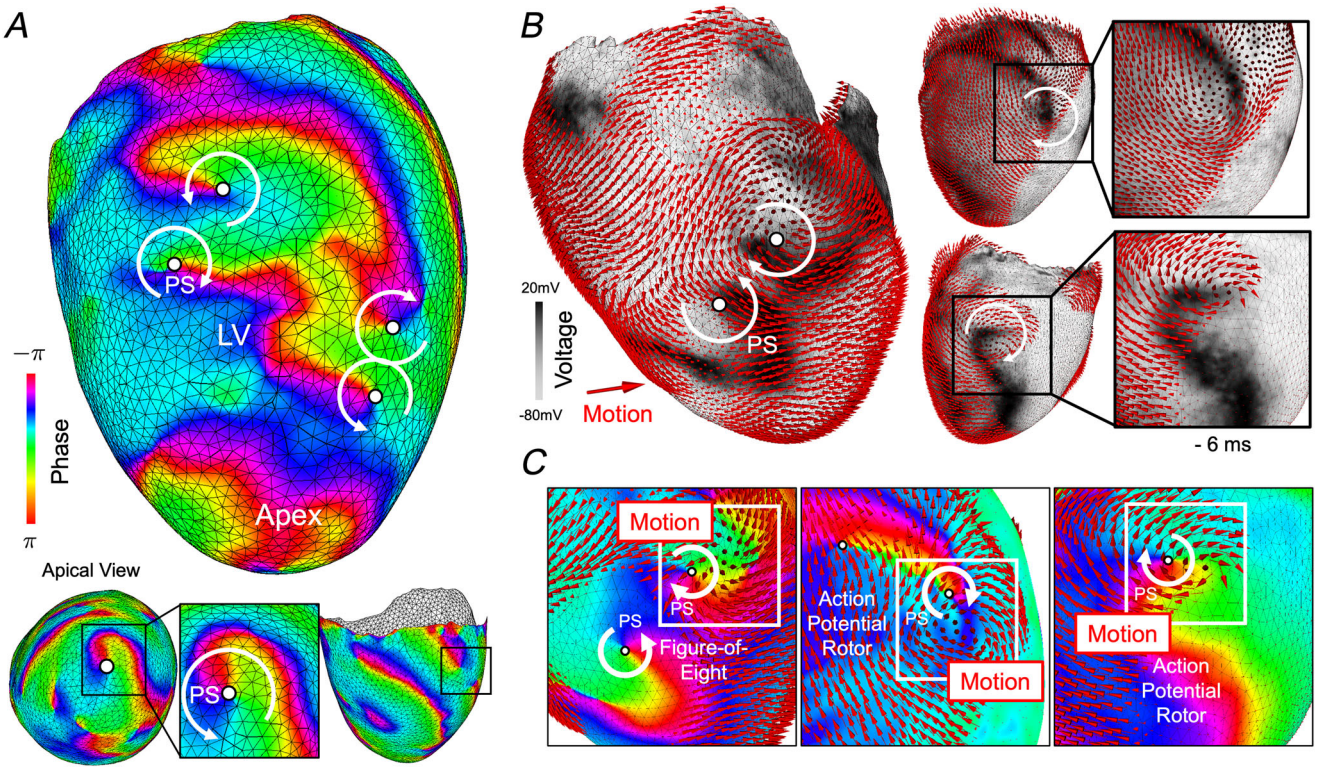


Figure 19. Vortex-like rotating action potential waves on the ventricular surface of an isolated fibrillating and contracting rabbit heart (see Supporting information, Videos SV18 and SV19)
A, phase maps showing action potential vortex waves including rotors with core regions marked by phase singularities (PS) in a rapidly fibrillating heart with little residual motion (VF1, same data as shown in Fig. 18. *B*, rotating action potential vortex waves (black) cause curl-like rotating mechanical deformation patterns (red vectors). To highlight curls in the displacement field, the vectors are normalized and therefore do not encode magnitude but only direction. Tissue displacements calculated with respect to tissue's mechanical configuration 20 ms earlier. Inserts show the same area 6 ms earlier. In some regions, the vectors point into the heart muscle and are therefore not visible. This episode (VF2) shows a more strongly contracting heart than in (*A*). *C*, phase maps of action potential vortex wave activity displaying phase singularities (PS) indicating rotational core region together with mechanical curls in the displacement vector fields, which emerge in close vicinity to electrical PS.

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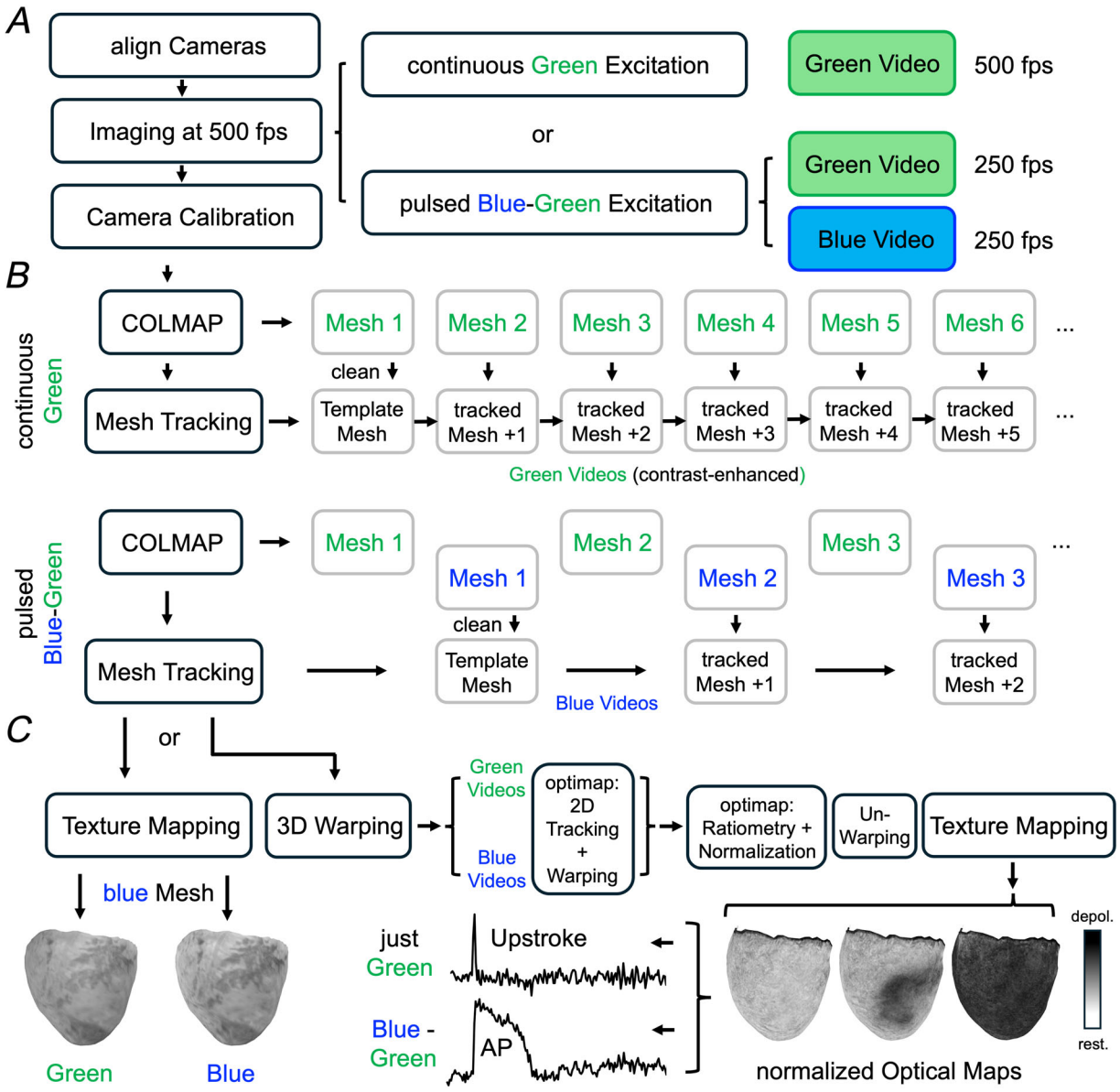


Figure 20. Schematic of workflow during imaging and post-processing (see Methods)
A, imaging was performed in either non-ratiometric mode with continuous green excitation (500 frames s^{-1}), or in ratiometric mode with pulsed blue-green excitation (effective frame rate 250 frames s^{-1}). The cameras were aligned before and calibrated after the imaging experiment. *B*, COLMAP was used to create static raw 3D meshes (see Fig. 4*A*). In continuous green mode, the first mesh of the green video sequence was used to create a template mesh (see Fig. 4*B*), which was then used to track tissue motion in the green (optional: contrast-enhanced) videos (right arrows). In pulsed blue-green mode, the first mesh of the blue video sequence was used to create a template mesh, which was then used to track tissue motion in the blue videos (right arrows). The tracking is constrained by the raw meshes (vertical arrows). *C*, 3D tracked meshes were either used to directly map raw (greyscale) video images (from the green or blue channels) as textures onto the 3D mesh (non-ratiometric mode: green, ratiometric mode: blue) or to post-process the videos to obtain normalized optical maps and texture-map these onto the 3D mesh. In the latter case, the 3D tracking data were used to stabilize/warp the videos (see Supporting information, Video SV7). Subsequently, residual motion was tracked and stabilized/warped using 2D tracking (using optimap) (see Supporting information, Video SV8). In ratiometric mode, 2D tracking was performed in both the green and blue channels, and the warped videos were subsequently combined into a ratiometric video. In non-ratiometric mode, the (pixel-wise) difference was computed between subsequent frames in the warped green videos. Lastly, the videos were (pixel-wise) normalized, unwarped using the 2D and 3D displacement data, and subsequently texture-mapped onto the moving 3D meshes. Accordingly, in continuous mode, the optical traces show the upstroke, and in ratiometric mode, the time-course of the action potential (AP). For further details, see Methods.

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3D numerical motion tracking. With our setup, we can measure electrical and mechanical phenomena simultaneously across the heart surface at high spatial and temporal resolutions. For example, we can measure electrical and mechanical activation times, APD, tissue strain and strain-rates at thousands of locations across the 3D heart surface. The measurement data are invaluable and can be used to study the heart's electromechanics in unprecedented detail. In the future, it could be used to study disease mechanisms and fundamental properties of cardiac excitation–contraction and metabolic physiology, or to inform and calibrate electromechanical computer models of the heart.

Sinus rhythm

Sinus rhythm poses the most challenging condition in optical mapping studies with contracting hearts. During sinus rhythm, the heart usually contracts and deforms more strongly, which makes the analysis more challenging in two specific ways: first, tracking the tissue and, second, measuring artifact-free optical traces of the action potential. Tracking is difficult particularly when the heart exhibits large translational or torsional movements. For example, when tracking with a single camera, the heart probably moves out of the field of view or rotates, such that 2D tracking algorithms fail to detect shifts of the tissue within the single camera view (see Supporting information, Video SV7. The second challenge lies in the strong back and forth movements of the heart relative to the excitation light, which can lead to local changes in the excitation of the fluorescent dye. This may, depending on the strength of the motion and the inhomogeneity of the excitation light, produce strong measurement artifacts and prohibit measuring the time-course of the action potential. We overcome these challenges with our 3D multiview mesh-based tracking approach combined with ratiometric voltage-sensitive imaging. The efficacy of our imaging approach is demonstrated by the quality of the optical action potential measurements across the strongly deforming heart surface (Figs 1*G*, 8*A* and 11; see also Supporting information, Videos SV3 and SV13). Our system enables us to measure APD (Figs 12*E* and 13) as well as APD and contractile changes, and directly relate them to each other (Figs 12*D*). In the future, this could allow more detailed studies of, for example, the long-QT syndrome, particularly with a focus on the relationship between repolarization, mechanics and arrhythmogenesis (Bekke et al., 2015), the effect of stretch-activated ion channels onto electrical propagation pathways, or electro-mechanical therapeutics, such as cardiac resynchronization therapy, among many other topics.

Electromechanical waves during pacing

Our data shown in Figs 15*C*–*F* and 16–17 provide further evidence for the high correlation between electrical and mechanical activation in the heart, and the first direct high-resolution observation of electromechanical waves. Previous studies described correlated electrical and mechanical activation or electromechanical waves (Christoph et al., 2018; de Boeck et al., 2008; Delhaas et al., 1993; Kroon et al., 2015; Lebert & Christoph, 2019; Maffessanti et al., 2020; Molavi Tabrizi et al., 2022; Provost et al., 2011; Wyman et al., 1999), but measurements never provided all data at once or at sufficiently high resolutions to resolve electromechanical waves. Studies provided either only mechanical data (de Boeck et al., 2008; Prinzen et al., 1999; Provost et al., 2011; Wyman et al., 1999), 2D epi- or endocardial electrical and mechanical data (Delhaas et al., 1993; Kroon et al., 2015), partly 2D epicardial electrical and 3D transmural mechanical data (Christoph et al., 2018), and/or low-resolution data of both electrical and mechanical activation (Delhaas et al., 1993; Kroon et al., 2015; Maffessanti et al., 2020). Kroon et al. (2015) and Maffessanti et al. (2020) assessed electromechanical activation in the ventricles using modified electroanatomical catheter mapping with integrated deformation measurements. Konofagou et al. coined the term 'electromechanical wave'. For example, Provost et al. (2011) imaged mechanical strain waves in patients *in vivo* at high-speed using ultrasound, referring to the technique as electromechanical wave imaging. These mechanical waves were then compared with simulated electrical waves generated by computer models, without providing simultaneous measurements of both the electrical and mechanical quantities that would enable a direct comparison. Christoph et al. (2018) provided optical mapping data of focal action potential wave spread together with corresponding 3D ultrasound data of focal strain patterns. Nevertheless, the ultrasound data were obtained at low spatial and temporal resolutions and not cross-registered, prohibiting the tracking of strain waves and aligning them precisely with electrical waves in space. Delhaas et al. (1993) provided data most similar to ours using an electrode brush and tracking optical markers recorded on video across the heart's surface. However, by comparison, our data provide much higher spatial resolutions, do not require cross-registration and cover the entire ventricles in 3D. Because of the high resolution, we can observe strain waves following electrical activation across both ventricles. Until now, such simultaneous high-resolution 3D observations of both the electrical and mechanical wave phenomena have been missing. Electromechanical waves were also described in cell cultures (Burton et al., 2015; Liu et al., 2023). Nevertheless, although cell cultures are an important tool for studying electromechanical

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waves in a more controllable setting, they are difficult to compare to *in vivo* or *ex vivo* data because the cells' movements are often restricted. In monolayer cell cultures in which cells are seeded onto a stiff substrate, mechanical waves reflect local cellular contractility and are not associated with large deformations and strains.

Our data show that the electrical and mechanical activation times do not correlate well everywhere across the ventricles (Fig. 16*E* and *F*; see also Supporting information, Video SV17). This is not surprising given that electrical and mechanical waves are fundamentally different physical phenomena arising in an excitable and elastic system. In general, we presume that there are many counter-examples where electrical and mechanical activation and wave speed do not match. The mechanical wave shown in Fig. 16*B* and *D* emerges during propagation along the fibre orientation. In this case, the contractile motion is aligned with the wavefront propagation direction. As previously described in the extended data comprising fig. 8 in Christoph et al. (2018), fig. 3 in Lebert and Christoph (2019) and figs 5 and 6 in Molavi Tabrizi et al. (2022), this configuration facilitates the mechanical wave phenomenon. The observation is also determined by how we define the mechanical activation time: the sudden transition from dilating to contracting strains is dependent on the alignment between the wave normal and local muscle fibre direction and can only be measured sufficiently far away from the pacing location. To observe the wave locally, the tissue must first dilate and then contract. The tissue surrounding the pulse's origin begins contracting almost immediately. However, the situation might be different in a different experimental setting (e.g. optical stimulation).

Our data shows that, although coupled electromechanical wave phenomena exist, mechanical wavefronts appear to be unreliable approximations of electrical wavefronts. Although they may be in good agreement, their relationship appears to be more complex, and further research is needed to better understand this relationship. Their degree of correlation probaly depends on various factors, such as mechanical pre- and after-load, atrial *vs.* ventricular activation, fibre anisotropy, wave propagation direction relative to the local fibre orientation, activation type and frequency, elastic inhomogeneities, and mechanical boundary conditions, as was also described in parts in a numerical study by Molavi et al. (2022). In general, inverse computational approaches (Christoph & Lebert, 2020; Lebert et al., 2023) or additional structural information (Giardini et al., 2025) could help to better identify electrical activation fronts from mechanical deformation or determine their degree of correlation. Further systematic study of electrical and mechanical activation in pre-clinical models and under various conditions (e.g. different pacing regimes) is needed to better understand these phenomena.

Electromechanical waves could be used for diagnostics not only as approximations of electrical activation patterns, but also because they may reflect electromechanical asynchrony in disease states. We observed electromechanical waves during pacing but not during sinus rhythm, probably because electrical activation is much faster and occurs from multiple sites simultaneously during the normal heartbeat. During ventricular pacing, the waves are slower and propagate from a singular origin towards a focal point where they merge. Accordingly, there is a region with quasi-planar wave spread in between, which presumably facilitates the observation of closely correlated electrical and mechanical waves. The epicardial measurements may not be adequate to observe electromechanical waves during sinus rhythm, or our imaging speeds (2 ms/4 ms) may not be sufficient to resolve them. We intend to study electromechanical waves more systematically and in more detail in future work.

Ventricular fibrillation

Our data shown in Fig. 19 (see also Supporting information, Video SV19) confirm previous observations of electromechanical vortices during VF that were made when analysing single-camera optical mapping data (Christoph et al., 2018). Here, we present, for the first time, corresponding 3D measurements of action potential vortex waves and co-localized vortex-like tissue deformation across the ventricular surface during VF. While previous observations characterized the vortices as electrical and mechanical phase singularities (Christoph et al., 2018), here, we present 3D displacement vector fields that exhibit a vortex-like curl structure. We intend to study these phenomena during VF with our imaging system more systematically in the future.

Practical considerations

Although we used 12 cameras in the present study, our imaging chamber fits up to 24 cameras. We anticipate that, in the future, we will use up to 24 cameras to image the whole heart, including the atria, and eventually employ two sets of cameras to image action potential and calcium waves. This would enable us to study the three main physiological players underlying the heartbeat: voltage, calcium and contraction. Our overall approach and design of the imaging setup utilize the recent availability of industrial-grade low-cost cameras (Lee et al., 2017). These cameras produce lower-quality videos, but averaging the overlapping video images across the 3D heart surface produces high-quality optical signals. The lower costs enabled us to integrate far more cameras into our setup than previously possible: in the past, multiple-camera

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optical mapping systems consisted of usually not more than four scientific cameras (Christoph et al., 2017; Zhang et al., 2016), the main limiting factor being cost. Although low-cost cameras were already used for panoramic optical mapping of non-moving hearts (Lee et al., 2017) and although three to four cameras are usually sufficient for mapping the entire surface of non-moving hearts (Bray et al., 2000; Kay et al., 2004, 2006; Rogers et al., 2007), it was estimated in Christoph et al. (2017) that at least eight cameras would be necessary to image the entire ventricular surface of a contracting heart.

In the present study, we showed that operating 12 or potentially more cameras is technically and practically feasible. We streamed the video data from 12 cameras onto a single computer using USB, and we did not experience major issues with camera failures, the high data rates or calibrating the cameras. Occasionally, cameras failed or could not be included during the data analysis, but this could be tolerated to some extent because one camera could usually be compensated by surrounding cameras if there was sufficient coverage of that area of the heart. Furthermore, the high data rates and large amounts of data that need to be captured, stored and processed during and after each experiment do pose a challenge. With 24 cameras and video image resolutions of 440 × 320 pixels, we would obtain data rates of three gigabytes per second. Accordingly, a 10-s long recording with 24 cameras would create 30 Gb of data. In our case with 12 cameras, a single experiment with 25–50 recordings produced on average about 0.5–1.0 Tb of data. Although we crop the video images and discard unnecessary data during the experiment to save storage space, further work is needed to compress the raw data into a suitable format for long-term storage.

Despite the many cameras, our imaging setup is cost-effective with an overall approximate cost of $20,000 for 24 cameras ($400 each) including the respective lenses ($500–800 each). By comparison, a single scientific camera costs $10,000–50,000 (e.g. Brainvision SciMedia MiCAM Ultima/N256, SciMedia Ltd, Costa Mesa, CA, USA; Teledyne Photometrics Evolve 128, Teledyne Vision Solutions, Chestnut Ridge NY, USA; Andor, Oxford Instruments, Belfast, UK) and a setup of four to 10 scientific cameras would correspondingly cost $100,000–$500,000, which is prohibitively expensive for most laboratories. Adopting our low-cost design with industrial-grade cameras could enable more researchers to venture into the world of panoramic 3D optical mapping. Costs for emission or excitation filters vary widely depending on the application ($30–50 per filter for standard non-ratiometric longpass emission filters *vs*. $500–1000 per filter for custom-made ratiometric bandpass filters; no excitation filters for non-ratiometric imaging and $100–200 per excitation filter for ratiometric imaging).

Advantages

The ability to image contracting hearts (with un-inhibited contraction) not only allows the exploration of interactions between the heart's electrophysiology and mechanics, but also, importantly, creates a more physiological *ex vivo* environment. This will enable novel optical mapping studies, in which the heart's behaviour and dynamics can be characterized in its more natural, beating state. When beating, the heart's oxygen and energy consumption is higher, which not only has implications regarding its metabolism, but also affects its electrophysiology (Garrot et al., 2017; Kappadan et al., 2020). Subsequently, our imaging platform can open new lines of research; for example, in the areas of cardiac metabolism, energetics or pH regulation (Lyu et al., 2022; Vaughan-Jones et al., 2006).

The two main technical advantages of our panoramic optical mapping approach are that it neither relies on 1) fiducial markers (Zhang et al., 2016) nor on 2) any additional shape reconstruction methods (Christoph et al., 2017). It therefore provides dynamic high-resolution 3D reconstructions immediately, and, in principle, it could also provide these reconstructions automatically and in real-time. As already mentioned in the introduction, the method described in Zhang et al. (2016) required fiducial markers, which impose lower spatial resolutions and may affect the tissue's mechanics. In the study by Christoph et al. (2017), it was necessary to perform a static shape reconstruction with the excitation–contraction uncoupling agent blebbistatin. This additional step made the approach impractical as it limited the time available for imaging to 5–15 min because the motion tracking and shape reconstructions could become compromised as a result of swelling of the heart or photo-bleaching or wash-out of the fluorescent dye. With our approach, there are no such constraints. We can perform measurements over 1–2 h, calibrate the cameras, track the tissue and generate 3D dynamic reconstructions in a fully automatic fashion. In principle, with further streamlining and optimization of the algorithms, the measurements could be performed close to or in real-time.

Another advantage is that the soccer ball-shaped imaging chamber allows us to image with up to 24 cameras and illuminate the heart from as many as 48 pre-defined locations. Both the cameras and light sources are evenly spaced and distributed around the heart. This circumvents several issues related to space restrictions, uneven illumination and excitation of the heart surface, as well as other complications that arose with previous experimental setups (Christoph et al., 2017, 2018). A crucial element in optical mapping studies with contracting hearts is to illuminate the heart as evenly as possible (Fig. 5*H*). Homogeneous illumination minimizes illumination-related motion artifacts, which are

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caused by relative motion between the tissue and the light sources; see the limitations section within the discussion in the study by Lebert et al. (2022). Although it may seem impossible to illuminate the heart perfectly homogeneously, our illumination scheme produces smaller illumination gradients than many other approaches. For example, in octagonal baths with eight glass walls, which were used in the studies by Christoph et al. (2017) and Christoph et al. (2018), the fewer windows restrict the positioning of both the cameras and light sources (e.g. LEDs). The cameras need to image perpendicularly through the windows, and the light sources need to be positioned in a way that avoids back-reflections into the camera images. Additionally, objects in the way of the light-sources (e.g. silicone seams between glass wall) can create shadows on the heart surface. Our chamber design allows the unobstructed, even illumination and imaging of the heart. A comparable approach using a spherical geometry for an optical fibre-based imaging setup was reported in Rieger et al. (2021).

Lastly, we achieved high spatial resolutions thanks to numerous cameras equipped with low-cost CMOS sensors, which produce images in the order of 300×300 pixels at 500 frames s^{-1}. We measured electrical wave phenomena on the entire ventricular surface with about 0.5–1 million pixels at spatial resolutions of about 120 μm per pixel. At these resolutions, we can identify single small vessels across the ventricular surface. Currently, optical mapping is typically performed only on parts of the ventricular surface, with video images ranging in the order 100×100 pixels with charge-coupled device cameras or 1000×1000 pixels with CMOS cameras. The sensor used in the present study comprises a lower dynamic range and smaller pixels (6.9×6.9 μm) associated with lower light-efficiency and signal-to-noise ratio (than e.g. Teledyne Photometrics Evolve 128) (Lee et al., 2017). However, we demonstrated that the signal quality is very high when averaging the signals from multiple cameras. In a previous study, we determined that the video quality of the low-cost cameras is sufficient for 2D tracking (Lebert et al., 2022).

Limitations

More work is necessary to establish robust atrial imaging. In our experience, imaging the contracting atria is more challenging than the ventricles because their geometry is more complex, featuring both convex and concave surfaces, and generally more uneven surfaces, which are more difficult to reconstruct and track in 3D. Furthermore, the atria contract more strongly and are often occluded by other tissue or parts of the atria (e.g. the appendages), with the latter aspect making the initial 3D reconstruction challenging. For atrial imaging to work robustly, we would use a different camera configuration and better 3D reconstruction techniques, which can handle complex 3D geometries and perform reconstructions despite occlusions and restricted views.

Measuring action potential waves is more challenging than measuring upstroke timings or electrical activation maps because tracking and ratiometric compensation must provide accurate optical traces over a longer period of time. Although the combination of our LED configuration with ratiometric imaging is effective in reducing illumination-related motion artifacts, residual inhomogeneities of the 3D light field inside the imaging chamber remain, and minuscule differences between the green and blue channels contribute to residual artifacts, diminishing the measurement quality, as seen in the traces in Figs 8*A*, 10 and 11. In future work, we aim to further minimize these inhomogeneities, also by adopting numerical approaches.

One challenging technical aspect during the experiments is associated with the low dynamic range of the cameras (12-bit: 4096 counts): we had to manually adjust the apertures and brightness of each of the cameras before and throughout the experiment to ensure that it would match across cameras and that the full dynamic range of each sensor was utilized. The final adjustments could only be made after staining and eventually needed to be repeated with each restaining. In addition to the aperture adjustments, we also occasionally had to adjust the brightness of some of the videos during post-processing to match the average brightness across the videos. This effectively minimizes artifacts during texture mapping, but a more automated approach would be desirable.

The accuracy of the 3D tracking algorithm is demonstrated indirectly by its effectiveness in reducing motion artifacts, which is noticeable in the optical traces (Figs 1*G*, 8*A*, 10 and 11), as well as in the optical maps (Figs 9*B* and 18*B*, *C*), the sharpness of the projected, averaged video images (see Supporting information, Video SV2) and the absence of optical flow in the texturized meshes (see Supporting information, Video SV5). Currently, we do not have means to directly validate the 3D tracking; for example, with synthetic data or a co-registered imaging modality serving as ground-truth. Simulations of a beating heart could be used to render synthetic optical mapping videos of the corresponding 3D scene, track the synthetic videos and compare the resulting moving 3D surface to the simulation, adopting a strategy similar to that used in previous 2D validation work (Christoph & Luther, 2018). Alternatively, the 3D reconstructions could be co-registered with high-resolution computed tomography (CT) scans of the heart.

Lastly, even though we used GPU-accelerated motion tracking, processing a 1.0 s long recording with 12 cameras

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currently takes several hours, excluding the time that is necessary to manually create a template mesh for tracking. We anticipate that the processing can be further streamlined and eventually be performed in real time in the future.

Conclusions

For over 30 years, optical mapping of action potential or calcium waves has been performed in contraction-inhibited hearts. As a result, studying the interplay between cardiac bioelectricity and biomechanics, comprising two of the main players involved in the heartbeat, has been limited. Here, we have demonstrated, for the first time, panoramic optical mapping of action potential waves in beating hearts. We imaged the electrical wave phenomena across the entire 3D deforming ventricular surface during normal and abnormal rhythms. Our optical mapping system provides high-resolution imaging data, which can be used to study both the electrical and mechanical physiology of the heart muscle in unprecedented detail, and could lead to new insights into the mechanisms of mechano-electric feedback.

References

Bachtel, A. D., Gray, R. A., Stohlman, J. M., Bourgeois, E. B., Pollard, A. E., & Rogers, J. M. (2011). A novel approach to dual excitation ratiometric optical mapping of cardiac action potentials with di-4-ANEPPS using pulsed LED excitation. *Institute of Electrical and Electronics Engineers Transactions on Bio-Medical Engineering*, **58**(7), 2120–2126.

Barnes, C., Shechtman, E., Finkelstein, A., & Goldman, D. B. (2009). PatchMatch: A randomized correspondence algorithm for structural image editing. *Association for Computing Machinery Transactions on Graphics (Proc SIGGRAPH)*, **28**(3), 1–11.

Ter Bekke, R. M. A., Haugaa, K. H., van den Wijngaard, A., Bos, J. M., Ackerman, M. J., Edvardsen, T., & Volders, P. G. A. (2015). Electromechanical window negativity in genotyped long-QT syndrome patients: Relation to arrhythmia risk. *European Heart Journal*, **36**(3), 179–186.

Blender-Online-Community. (2018). *Blender – a 3D Modelling and Rendering Package*. Blender Foundation. http://www.blender.org.

Bourgeois, E. B., Bachtel, A. D., Rogers, J. M., Huang, J., & Walcott, G. P. (2011). Simultaneous optical mapping of transmembrane potential and wall motion in isolated, perfused whole hearts. *Journal of Biomedical Optics*, **16**(09), 1.

Bray, M.-A., Shien-Fong Lin, & Wikswo, J. P. (2000). Three-dimensional surface reconstruction and fluorescent visualization of cardiac activation.″ *Institute of Electrical and Electronics Engineers Transactions on Biomedical Engineering*, **47**(10), 1382–1391.

Burton, R. A. B., Klimas, A., Ambrosi, C. M., Tomek, J., Corbett, A., Entcheva, E., & Bub, G. (2015). Optical control of excitation waves in cardiac tissue. *Nature Photonics*, **9**(12), 813–816.

Chen, Z., Nguyen, K., Kowalik, G., Shi, X., Tian, J., Doshi, M., Alber, B. R., Guan, X., Liu, X., Ning, X., Kay, M. W., & Lu, L. (2023). Transparent and stretchable Au–Ag nanowire recording microelectrode arrays. *Advanced Materials Technologies*, **8**(10), 2201716.

Christoph, J., Chebbok, M., Richter, C., Schröder-Schetelig, J., Bittihn, P., Stein, S., Uzelac, I., Fenton, F. H., Hasenfuß, G., Gilmour, R. F., & Luther, S. (2018). Electromechanical vortex Filaments during cardiac fibrillation. *Nature*, **555**(7698), 667–672.

Christoph, J., & Luther, S. (2018). Marker-free tracking for motion artifact compensation and deformation measurements in optical mapping videos of contracting hearts. *Frontiers in Physiology*, **9**, 1483.

Christoph, J., Schröder-Schetelig, J., & Luther, S. (2017). Electromechanical Optical mapping. *Progress in Biophysics and Molecular Biology*, **130**: 150–169.

Christoph, J., & Lebert, J. (2020). Inverse mechano-electrical reconstruction of cardiac excitation wave patterns from mechanical deformation using deep learning. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, **20**(12), 123–134.

de Boeck, B. W., Kirn, B., Teske, A. J., Hummeling, R. W., Doevendans, P. A., Cramer, M. J., & Prinzen, F. W. (2008). Three-dimensional mapping of mechanical activation patterns, contractile dyssynchrony and dyscoordination by two-dimensional strain echocardiography: Rationale and design of a novel software toolbox. *Cardiovascular Ultrasound*, **6**(1), 22.

Delhaas, T., Arts, T., Prinzen, F., & Reneman, R. S. (1993). Relation between regional electrical activation time and subepicardial fiber strain in the canine left ventricle. *Pflugers Archiv: European Journal of Physiology*, **423–423**(1–2), 78–87.

Efimov, I. R., Nikolski, V. P., & Salama, G. (2004). Optical imaging of the heart. *Circulation Research*, **95**(1), 21–33.

Farnebäck, G. (2003). Two-frame motion estimation based on polynomial expansion. In G. Goos, J. Hartmanis, J. van Leeuwen, J. Bigun, & T. Gustavsson (Eds.), *Image analysis* (vol. **2749**). Springer Berlin Heidelberg.

Fullenkamp, D. E., Maeng, W.-Y., Oh, S., Luan, H., Kim, K. S., Chychula, I. A., Kim, J.-T., Yoo, J.-Y., Holgren, C. W., Demonbreun, A. R., George, S., Li, B., Hsu, Y., Chung, G., Yoo, J., Koo, J., Park, Y., Efimov, I. R., Mcnally, E. M., & Rogers, J. A. (2024). "Simultaneous electromechanical monitoring in engineered heart tissues using a Mesoscale framework." *Science Advances*, **10**(37), eado7089.

Garrido-Jurado, S., Muñoz-Salinas, R., Madrid-Cuevas, F. J., & Marín-Jiménez, M. J. (2014). Automatic generation and detection of highly reliable fiducial markers under occlusion. *Pattern Recognition*, **47**(6), 2280–2292.

Garrott, K., Kuzmiak-Glancy, S., Wengrowski, A., Zhang, H., Rogers, J., & Kay, M. W. (2017). K(ATP) channel inhibition blunts electromechanical decline during hypoxia in left ventricular working rabbit hearts. *The Journal of Physiology*, **595**(12), 3799–3813.

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Giardini, F., Olianti, C., Marchal, G. A., Campos, F., Romanelli, V., Steyer, J., Madl, J., Piersanti, R., Arecchi, G., Perumal Vanaja, I., Biasci, V., Rog-Zielinska, E. A., Nesi, G., Loew, L. M., Cerbai, E., Chelko, S. P., Regazzoni, F., Loewe, A., Bishop, M. J., Mongillo, M., Kohl, P., Zaglia, T., Zgierski-Johnston, C. M., & Sacconi, L. (2025). Correlative imaging integrates electrophysiology with three-dimensional murine heart reconstruction to reveal electrical coupling between cell types. *Nature Cardiovascular Research*, **4**(11), 1466–1486.

Gray, R. A., Jalife, J., Panfilov, A., Baxter, W. T., Cabo, C., Davidenko, J. M., & Pertsov, A. M. (1995). Nonstationary vortexlike reentrant activity as a mechanism of polymorphic ventricular tachycardia in the isolated rabbit heart. *Circulation*, **91**(9), 2454–2469.

Iyer, A. N., & Gray, R. A. (2001). An experimentalist's approach to accurate localization of phase singularities during reentry. *Annals of Biomedical Engineering*, **29**(1), 47–59.

Kappadan, V., Sohi, A., Parlitz, U., Luther, S., Uzelac, I., Fenton, F., Peters, N. S., Christoph, J., & Ng, F. S. (2023). Optical mapping of contracting hearts. *The Journal of Physiology*, **601**(8), 1353–1370.

Kappadan, V., Telele, S., Uzelac, I., Fenton, F., Parlitz, U., Luther, S., & Christoph, J. (2020). High-resolution optical measurement of cardiac restitution, contraction, and fibrillation dynamics in beating vs. blebbistatin-uncoupled isolated rabbit hearts. *Frontiers in Physiology*, **11**. 464.

Kay, M. W., Amison, P. M., & Rogers, J. M. (2004). Three-dimensional surface reconstruction and panoramic optical mapping of large hearts. *Institute of Electrical and Electronics Engineers Transactions on Biomedical Engineering*, **51**(7), 1219–1229.

Kay, M. W., Walcott, G. P., Gladden, J. D., Melnick, S. B., & Rogers, J. M. (2006). Lifetimes of epicardial rotors in panoramic optical maps of fibrillating swine ventricles. *American Journal of Physiology-Heart and Circulatory Physiology*, **291**(4), H1935–H1941.

Khwaounjoo, P., Rutherford, S. L., Svrcek, M., Legrice, I. J., Trew, M. L., & Smaill, B. H. (2015). Image-based motion correction for optical mapping of cardiac electrical activity. *Annals of Biomedical Engineering*, **43**(5), 1235–1246.

Klaudiny, M., & Hilton, A. (2011). Cooperative patch-based 3D surface tracking. (*2011) Conference for Visual Media Production*, November. https://doi.org/10.1109/cvmp.2011.14

Kroon, W., Lumens, J., Potse, M., Suerder, D., Klersy, C., Regoli, F., Murzilli, R., Moccetti, T., Delhaas, T., Krause, R., Prinzen, F. W., & Auricchio, A. (2015). In vivo electromechanical assessment of heart failure patients with prolonged QRS duration. *Heart Rhythm*, **12**(6), 1259–1267.

Kwon, Y.-H. (1999). Object plane deformation due to refraction in two-dimensional underwater motion analysis. *Journal of Applied Biomechanics*, **15**(4), 396–403.

Lebert, J., & Christoph, J. (2019). Synchronization-based reconstruction of electromechanical wave dynamics in elastic excitable Media. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, **29**(9), 093117.

Lebert, J., Ravi, N., Fenton, F. H., & Christoph, J. (2021). "Rotor localization and phase mapping of cardiac excitation waves using deep neural networks." *Frontiers in Physiology*, **12**, 782176.

Lebert, J., Ravi, N., Kensah, G., & Christoph, J. (2022). "Real-time optical mapping of contracting cardiac tissues with GPU-accelerated numerical motion tracking." *Frontiers in Cardiovascular Medicine*, **9**, 787627.

Lebert, J., & Christoph, J. (2023). *Optimap: An open-source library for the processing of fluorescence video data*. https://doi.org/10.5281/ZENODO.8336455

Lebert, J., Deng, D., Fan, L., Lee, L. C., & Christoph, J. (2023). Deep learning-based prediction of electrical arrhythmia circuits from cardiac motion: An In-silico study, *arXiv:2305.07822*. https://arxiv.org/abs/2305.07822

Lee, P., Calvo, C. J., Alfonso-Almazán, J. M., Quintanilla, J. G., Chorro, F. J., Yan, P., Loew, L. M., Filgueiras-Rama, D., & Millet, J. (2017). Low-cost optical mapping systems for panoramic imaging of complex arrhythmias and drug-action in translational heart models. *Scientific Reports*, **7**(1), 43217.

Li, X., Almeida, T. P., Dastagir, N., Guillem, M. S., Salinet, J., Chu, G. S., Stafford, P. J., Schlindwein, F. S., & Ng, G. A. (2020). Standardizing single-frame phase singularity identification algorithms and parameters in phase mapping during Human atrial fibrillation. *Frontiers in Physiology*, **11**(July), 869.

Liu, W., Han, J. L., Tomek, J., Bub, G., & Entcheva, E. (2023). Simultaneous widefield voltage and dye-free optical mapping quantifies electromechanical waves in Human induced pluripotent stem cell-derived cardiomyocytes. *American Chemical Society Photonics*, **10**(4), 1070–1083.

Lyu, Y., Thai, P. N., Ren, L., Timofeyev, V., Jian, Z., Park, S., Ginsburg, K. S., Overton, J., Bossuyt, J., Bers, D. M., Yamoah, E. N., Chen-Izu, Y., Chiamvimonvat, N., & Zhang, X.-D. (2022). Beat-to-Beat dynamic regulation of intracellular pH in cardiomyocytes. *Iscience*, **25**(1), 103624.

Maffessanti, F., Jadczyk, T., Kurzelowski, R., Regoli, F., Caputo, M. L., Conte, G., Gołba, K. S., Biernat, J., Wilczek, J., Dąbrowska, M., Pezzuto, S., Moccetti, T., Krause, R., Wojakowski, W., Prinzen, F. W., & Auricchio, A. (2020). The influence of scar on the spatio-temporal relationship between electrical and mechanical activation in heart failure patients. *EP Europace*, **22**(5), 777–786.

Molavi Tabrizi, A., Mesgarnejad, A., Bazzi, M., Luther, S., Christoph, J., & Karma, A. (2022). Spatiotemporal organization of electromechanical phase singularities during high-frequency cardiac arrhythmias. *Physical Review X*, **12**(June), 021052.

Myles, R. C., Wang, L., Bers, D. M., & Ripplinger, C. M. (2015). Decreased inward rectifying k^+ current and increased ryanodine receptor sensitivity synergistically contribute to sustained focal arrhythmia in the intact rabbit heart. *The Journal of Physiology*, **593**(6), 1479–1493.

Narayan, S. M., & John, R. M. (2024). Advanced electroanatomic mapping: Current and emerging approaches. *Current Treatment Options in Cardiovascular Medicine*, **26**(4), 69–91.

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Nash, M. P., Mourad, A., Clayton, R. H., Sutton, P. M., Bradley, C. P., Hayward, M., Paterson, D. J., & Taggart, P. (2006). Evidence for multiple mechanisms in Human ventricular fibrillation. *Circulation*, **114**(6), 536–542.

Batchelor, O. (2023). Multical. https://github.com/Oliver-Batchelor/Multical.

Prinzen, F. W., Hunter, W. C., Wyman, B. T., & McVeigh, E. R. (1999). Mapping of regional myocardial strain and work during ventricular pacing: Experimental study using magnetic resonance imaging tagging. *Journal of the American College of Cardiology*, **33**(6), 1735–1742

Provost, J., Nguyen, V. T.-H., Legrand, D., Okrasinski, S., Costet, A., Gambhir, A., Garan, H., & Konofagou, E. E. (2011). Electromechanical wave imaging for arrhythmias. *Physics in Medicine and Biology*, **56**(22), L1–L11.

Rieger, M., Dellenbach, C., Vom Berg, J., Beil-Wagner, J., Maguy, A., & Rohr, S. (2021). Enabling comprehensive optogenetic studies of mouse hearts by simultaneous opto-electrical panoramic mapping and stimulation. *Nature Communications*, **12**(1), 5804.

Rodriguez, M. P., & Nygren, A. (2015). Motion estimation in cardiac fluorescence imaging with scale-space landmarks and optical flow: A comparative study. *Institute of Electrical and Electronics Engineers Transactions on Bio-Medical Engineering*, **62**(2), 774–782.

Rogers, J. M., Walcott, P. G., Gladden, J. D., Melnick, S. B., & Kay, M. W. (2007). Panoramic optical mapping reveals continuous epicardial reentry during ventricular fibrillation in the isolated swine heart. *Biophysical Journal*, **92**(3), 1090–1095.

Rohde, G. K., Dawant, B. M., & Lin, S.-F. (2005). Corrections of motion artifacts in cardiac optical mapping using image registration. *Institute of Electrical and Electronics Engineers Transactions on Bio-Medical Engineering*, **52**(2), 338–341.

Salama, G., Lombardi, R., & Elson, J. (1987). "Maps of optical action potentials and NADH fluorescence in intact working hearts." *American Journal of Physiology-Heart and Circulatory Physiology*, **252**(2), H384–H394.

Schönberger, J. L., Zheng, E., Pollefeys, M., & Frahm, J.-M. (2016). Pixelwise view selection for unstructured multi-view stereo. *European Conference on Computer Vision (ECCV)*.

Schroeder, W., Martin, K., & Lorensen, B. (2006). *The visualization toolkit* (4th ed.). Kitware.

Svrcek, M., Rutherford, S., & Smaill, B. (2009). Characteristics of motion artifacts in cardiac optical mapping studies. *Conference Proceedings, 31st Annual International Conference of the IEEE EMBS, Minneapolis, USA*, **31**, 3240–3243. https://doi.org/10.1109/IEMBS.2009.5333531.

Swift, L. M., Kay, M. W., Ripplinger, C. M., & Posnack, N. G. (2021). Stop the beat to see the rhythm: Excitation-contraction uncoupling in cardiac research. *American Journal of Physiology-Heart and Circulatory Physiology*, **321**(6), H1005–H1013.

Vaughanjones, R., Spitzer, K., & Swietach, P. (2006). Spatial aspects of intracellular pH regulation in heart muscle. *Progress in Biophysics and Molecular Biology*, **90**(1–3), 207–224.

Wyman, B. T., Hunter, W. C., Prinzen, F. W., & McVeigh, E. R. (1999). Mapping propagation of mechanical activation in the paced heart with MRI tagging. *American Journal of Physiology – Heart and Circulatory Physiology*, **276**(3), H881–H891.

Zhang, H., Iijima, K., Huang, J., Walcott, G. P., & Rogers, J. M. (2016). Optical mapping of membrane potential and epicardial deformation in beating hearts. *Biophysical Journal*, **111**(2), 438–451.

Zhang, H., Patton, H. N., Wood, G. A., Yan, P., Loew, L. M., Acker, C. D., Walcott, G. P., & Rogers, J. M. (2023). Optical mapping of cardiac electromechanics in beating in vivo hearts. *Biophysical Journal*, **122**(21), 4207–4219.

Zhou, Q.-Y.i, Park, J., & Koltun, V. (2018). Open3D: A modern library for 3D data processing. *arXiv:1801.09847*.

Additional information

Data availability statement

The data generated in our study shown in Figs 6A,B,C, 7A,B, 12A,B,C, 18 and 19 is available online (https://cardiacvision.ucsf.edu/videos/3d-optical-mapping).

Competing interests

The authors declare that they have no competing interests.

Author contributions

S.C. developed the numerical tracking and reconstruction algorithms. S.D. and J.L. assisted with developing and refining the algorithms. S.C., J.L. and J.C. analysed the data. J.L. and J.C. performed the experiments. S.C. and M.M. assisted during the experiments. J.L. and J.C. designed and built the imaging setup. J.L. developed hardware and software for imaging and data acquisition. J.L. established the barium chloride protocol. J.C. designed and built the imaging chamber, developed the ratiometric imaging and performed the corresponding data analysis. M.M. contributed to the validation and data analysis of the ratiometric imaging. C.G. developed the LED driver. J.C. designed the figures and wrote the manuscript. J.L. and J.C. conceived the research.

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Keywords

cardiac biomechanics, cardiac mapping, excitation–contraction coupling, mechano-electric feedback, motion tracking, optical mapping, photogrammetry, ventricular fibrillation, voltage-sensitive fluorescence imaging

Supporting information

Additional supporting information can be found online in the Supporting Information section at the end of the HTML view of the article. Supporting information files available:

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